



Master Thesis report

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Title: Feasibility study of Gisuma micro- hydropower plant(Rwanda).



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I.ABSTRACT

Nowadays; Rwanda is one of the country which is under developing country and one of the strategy to be used is to explore and develop the sector of energy in Rwanda.

Our country Rwanda needs to develop the rural electrification sectors in order to provide illumination to the population living in those rural areas for rapid development in different sectors like schools; health centers; business centers; administrative officers. As MSE we have this opportunity to show our contribution by making a feasibility study of Gisuma Micro-hydropower plant in Gisagara District.

A micro hydropower is used in the rural electrification and does not necessary supply electricity to the national grid. Micro hydro powers plants are utilized in isolated and off-grid like Gisuma areas.

In Rwanda the existing electricity network necessity high transmission lines and low load factor and this has an implication of high cost of extending to grid extension. The future Gisuma micro hydropower plant will be managed by the local population people of Gisuma sector and we hope to have a rapid change and development of daily life from Gisuma people.

To get all required information one of method to be used is the interview of peoples which are living in that region and site visits by measurement of important data such as head, discharge, topographic data of the river are presented. Then, the results obtained were analyzed using RETScreen software to demonstrate the technical feasibility of the plant to identify electromechanical equipments such as turbine, alternator, etc., to be used in that micro-hydropower.

II. PREFACE

Many people around the world live in areas where the water streams and Rivers are potential sources of energy supply for lighting, communication and processing industries (small and big). This has proven to be a very valuable natural resource, which can be exploited even at lower levels through building of small hydro power schemes that can go as low as few kilowatts to assist communities.

This project is to contribute among the existing other projects in rural development planning and improved energy supply through the simplest technology, “micro hydropower” that can be afforded and handled by rural villagers in Rwanda like Gisuma river taken as the case study in our project.

We have designed the flow of the chapters to be clear and logical, in order to ease the understanding of the feasibility study of Gisuma Micro-hydropower plant. We embrace both traditional and non-traditional fundamental concepts and principles of a Micro-hydropower plant, and the book is organized as follows:

In Chapter 1, We introduce the problem of lack of sufficient power generation capacity in Rwanda. To solve this crisis problem, the projects of identifications of all resources have been started especially in micro hydro power which is available around our country.

In Chapter 2, We review the Background Rwanda Country policies to know energy situation in Rwanda and background on hydropower concept to know the requirement of a micro-hydropower plant.

In Chapter 3, we study about the Hydrology and Geology of Gisuma region to gather data required for the evaluation of the energy production and techniques used to get data in the micro-hydropower scheme which is the mean daily flow series at the scheme water intake in a period that has to be long enough in order to represent, in average, the natural flow regime.

In Chapter 4, we study about the environmental impact assessment and its identification applied to Gisuma micro-hydro power plant which consists in evaluation of the favorable and unfavorable impacts in natural and social environmental context implications.

In Chapter 5, we study the conceptual design of Gisuma micro-hydropower plant where this section presents the details of various alternative calculations of available power and calculation of different size of different equipments (penstock diameter, etc,...) necessary to be used and the estimation of different electromechanical equipments(hydro turbine, generator, transformer, etc,...) necessary related to the obtained results considered during the reconnaissance survey at the proposed site. Out of the numerous alternatives the major alternatives that may be feasible and worth considering are discussed in this chapter.

In Chapter 6, we study the economic analysis in terms of investment in Gisuma Micro-hydropower scheme will incur costs as well as earn income over the life of the project.

In Chapter 7, we provide the recommendations and advices in order to implement successfully the project.

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List of Abbreviations

AC: Alternating Current

AE: Actual Evapotranspiration

CAD: Canadian Dollars

CO₂: Carbon Dioxide

DC: Direct Current

EDPRS: Economic Development and Poverty Reduction Strategy

EIA: Environmental Impact Assessment

ESHA: European Small Hydropower Association

EWSA : Energy and Water Sanitation Authority

EWSA: Energy Water and Sanitation Authority

GDP: Growth Domestic Product

GHG: Greenhouse Gas

GoR: Government of Rwanda

GWP: Global Warming Potential

HDPE: High Density Poly Ethylene

Hz: Hertz

IPPs: Independent Power Producers

JICA:Japan International Cooperation Agency.

KIST: Kigali Institute of Science and Technology

KV: Kilovolt

kW: Kilowatt

KWh: Kilowatt-hour

LV: Low voltage

MDGs: Millennium Development Goals

MHP: Micro hydro Power

MINALOC: Ministry of Local Government

MINEDUC: Ministry of Education

MINICOFIN: Ministry of commerce and Finance

MINICOM: Ministry of Commerce

MININFRA: Ministry of Infrastructure

MINIRENA: Ministry of Natural Resources

MV: Medium voltage

MW: Mega-watt

NO_x : Nitrous Oxide

NUR: National University of Rwanda

PAT: Pump As Turbine

PE : Potential Evapotranspiration

pH: Potential Hydrogen Concentration

REMA: Rwanda Environment and Management Agency

RETs: Renewable Energy Technologies

RHS: Right Hand Side

RNC: Natural Resources Canada

RPM: Revolutions per Minute

RURA: Rwanda Utilities Regulatory Agency

RWf: Rwanda francs

SACCO: Saving and Credit Cooperative

SHP: Small Hydro Power Plant

SO₂: Sulphur Dioxide

SWA_p : Sector Wide Approach

TSS: Total Suspended Solids

USD: United States Dollars

Chapter 1.INTRODUCTION

Rwanda, with its emerging commerce and industries, is facing a daunting task to cope up with the power crisis. There is a lack of sufficient power generation capacity, and the existing national grid network is unable to power the whole nation. The rural and remote areas have a low-load demand but the electricity supply has been characterized by high transmission and distribution costs, transmission losses, and heavily subsidized pricing. The demand for power is increasing at a rapid pace although the generation of power has not increased at the same proportion. The gap between demand and supply of power is quite significant. The shortage of power generation capacity is estimated to be around 100 MW and about 16.5% of households are connected to the grid.

To solve this crisis problem, the projects of identifications of all resources have been started especially in micro hydro power which is available around our country. In this way Gisuma river (which is located in Gisagara District in south province near Burundi border) has been chosen as source of micro hydropower to be based on by making feasibility study of hydropower plant of our project. In this paper a summary of what we have done are shown here.

1.1. Project Description

Generally, a micro-hydro power plant is a system where one can produce as much as a few hundreds of kilo-Watts of electricity using the run-of-river source. This method is used in the regions where there is availability of different river sources for hydroelectric power, but at low discharges so that one can produce about 451kW. This is mostly applicable in areas, as it is currently the case in some Southern parts of Rwanda where there is presently inaccessibility to electricity from the national grid.

In this study, we used the design and working principles of micro hydro power plants to generate power from the Gisuma River, which has its upper catchments in the mountains of Muganza hill.

This project can be used to install about 451kW capacity hydro-power plant and it is to be developed as an off- grid hydro power project supplying electricity for a small isolated off-grid village setting.



Figure 1. Location of Gisuma River on country map. (Source REMA, 2009)



Figure 2. Gisuma River region (GPS photo, 2013).

Rwanda is a developing country, which presents large rural and isolated regions without access to grid electricity. This situation, presents some difficulties to the population, who suffer from lack of infrastructure like easy accessibility to electricity in the clinics, primary schools, colleges and commercial centers.

It is hoped that the development of proposed Gisuma micro hydro power project, will be financially and technically feasible, and an infrastructure that would positively impact the social, productive and living standards of people in the Muganza sector of Rwanda. Furthermore, the feasibility of this project will also be contributing to the country's electrification level and rural development, which bolster the efforts of the Government of Rwanda on the way to becoming a developed nation .

1.2. Project location

River Gisuma originates in the Muganza mountain ranges of southern Rwanda. These rivers flow through mountains and into a steep slope at the site in the Gisuma catchment. All these conditions favor efficient and economical water resources development.

The proposed project is located very close to a district road and therefore the transportation of the materials will be convenient and the cost of construction will be lower. The scheme intends to utilize the natural drop of river elevation to generate hydropower with the available discharge. The river flow is diverted at the top of the natural rapids and after generation, the water is discharged back to the same river at a downstream location at the end of the natural drop before used in irrigation purpose from the proposed power house.

The scheme will have an installed capacity of 451 kW using $0.568 \text{ m}^3/\text{s}$ and a net head of 125 m. This will function as an off-grid electricity generation scheme.

The proposed project has minimal adverse effects on the environment and the mitigating measures of any possible adverse effects are discussed later in this project report.

The geological study of the area will be done before implementation for providing better information about the foundation quality, which should be adequate for the stability of the civil engineering structures, the weir, canal, penstock and the powerhouse.

1.3. Project Implementation

The construction of Gisuma micro- hydro power plant with proposed power output capacity of 451 kW and installation of the local mini-grid and distribution lines to the targeted consumers (households, social services, local institutions and schools) would be charged on monthly basis and should last for two years . Billing will be according to the loads connected to the mini-grid, independent metering will be installed to each commercial customer so as to simplify monthly electricity revenue collection in accordance with electricity usage. The revenue accrued from customers will be used for maintenance of the plant and village development activities.

This project activity also contemplates the production of clean power that will contribute to reduce dependence on imported kerosene which is more used in the region for house lighting and reduce greenhouse gases emission specifically CO₂, which would have occurred otherwise, in the absence of this project.

Chapter 2. BACKGROUND

2.1. Country Background

Rwanda is a country in central and eastern Africa with a population of approximately 11.4 million (2011) on total size of 26,338 square kilometers with 433 inhabitants per km². Rwanda is among the highest population density in Africa ²¹.

Rwanda is a landlocked country bordered by the Democratic Republic of Congo from west, Uganda; north, Tanzania; east and Burundi to the south. The entire country is located at high altitude; the lowest point is Rusizi River at 950m above sea level ²².

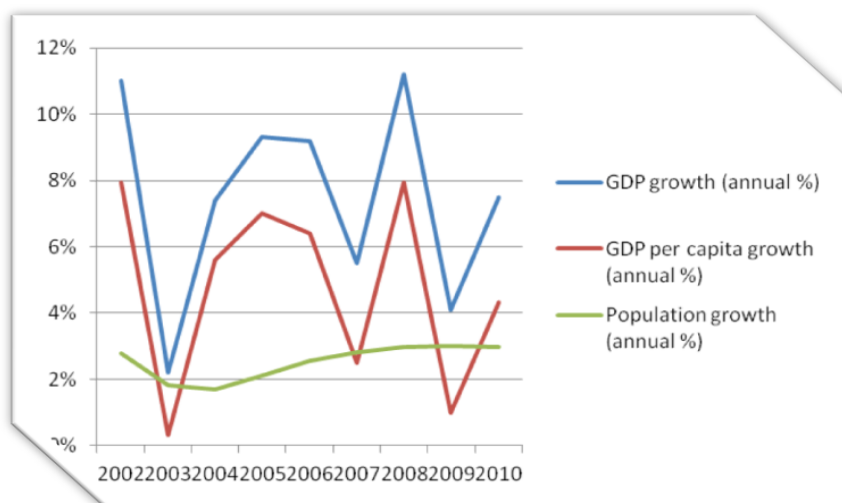


Figure 4. GDP (Growth Domestic Product) growth of Rwanda ^[9]

Rwanda is a country of few natural resources and the economy is based mostly on subsistence agriculture by local farmers using simple tools ⁴. An estimated 90% of the working citizens rely on subsistence farming with an estimate of 42.1% of GDP by 2010 ^[4].

Rwanda has a small-sized economy, but with one of the fastest rates of development in East and Central Africa. The government of Rwanda recognizes the key role of the private sector in accelerating growth and eradicating poverty, and straggling for innovative ways to finance its development beyond traditional partners and instruments. It has accordingly undertaken reforms to improve the business environment and to reduce the cost of doing business. Rwanda was named top performer in the 2010 Doing Business report ²¹(NISR, National Institute of Statistics of Rwanda. 2010), among the 10 most improved economies in 2011, and ranked third easiest place to do business in Africa in 2012. Rwanda's economic outlook for 2012 is positive, but with increasing medium-term risks. Real GDP is projected to slow down in 2012 and further more in 2013 and 2014, due to the impact of fiscal consolidation efforts and the uncertainties of the global economic outlook ¹⁷.

2.2. Country policies

Rwanda faces significant challenges to meeting access to clean energy and its targeted goal is to increase access to modern energy to meet the power demand for economic development of the country, which this thesis work aims to address. Rwanda's electricity sector is effective by regional standards, but progress in generation and access clean energy needs to speed up for the goals to be met.

2.2.1. Energy in Rwanda⁸

Electricity accounts for only about 5% of primary energy use in Rwanda. Biomass is the primary source of energy accounting for some 84% of primary energy use, and petroleum products account for the rest. Rwanda has one of the lowest electricity consumption per capita compared to other countries in the region, and generation capacity is low –the country currently has about 100. MW of installed capacity and only about 11% of households are connected to the grid.

The existing installed generation capacity and available capacity is show in Table 1.

Of the installed generation capacity, hydropower accounts for about 59%, thermal generation, primarily hired diesel and heavy oil fuel based generation units, for 40%, and methane gas for about 1%.The high reliance on thermal generation comes at a significant cost to Rwanda, especially given the present high prices for oil products.

Table 2 Current situation of Energy in Rwanda

No	Plant Name	Type	Year of Operation	Installed Capacity in MW
1	Mukungwa	Hydro	1982	12
2	Ntaruka	Hydro	1959	11.25
3	Gisenyi	Hydro	1957	1.2
4	Gihira	Hydro	1984	1.8
5	Jabana 1	Thermal	2004	7.2
6	Aggreko Gikondo	Thermal	2005	10
7	Aggreko Mukungwa	Thermal	2012	10
8	Jali Solar	PV	2007	0.25
9	K P 1	Thermal	Nov,2008	3.6
10	Jabana 2	Thermal	Feb,2009	21
11	Murunda	Hydro	Apri,2010	0.1
12	Rukarara	Hydro	Dec,2010	9
13	Rugezi	Hydro	Mar,2011	2.2
14	Keya	Hydro	July,2011	2.2
15	Nkora	Hydro	July,2011	0.68
16	Cymbili	Hydro	July,2011	0.3
17	Mukungwa II	Hydro	Apri,2013	2.2
18	Mazimeru	Hydro	May,2012	0.5
	Total			95.48
	Total Installed Generation Capacity	112.48	MW	

Source: Electricity Development Strategy 2011-2017, MININFRA, March 2011

2.2.2. ENERGY LAWS, POLICIES, AND ENABLING ENVIRONMENT

2.2.2.1. Electricity Law

Rwanda's draft Electricity Law was enacted into law in June 2011 and gazette in July 2011.

The law on electricity governs the activities of, electric power, production transmission distribution and trading both within and outside the national territory of Rwanda. The primary objectives of the law are:

- Liberalization and Regulation of electricity sector;
- Harmonious development of power supply for all population categories and for all the Country's economic and social development sectors in the framework of laws in force;
- Setting up economic conditions enabling electric power sector investments;
- Respect for the conditions of fair and loyal competition and for rights of users and operators.

The Electricity Law gives the Ministry in-charge of electricity the rights to provide concession Agreements to firms, and provides the legal basis for the Rwanda

Utilities Regulatory Agency (RURA) to approve and grant licenses for the production, transmission, distribution and sale of electricity, the conditions for licensing, and addresses the rights and obligations of the license holders.

The Law specifies that the electricity market of Rwanda shall be a single market based on free and open third party access to the transmission and distribution networks based upon the principles of regulated access to ensure a transparent and non-discriminatory

Market place.

The Electricity Law authorizes the issuance of an International Trade License for the import and export of electric power across the borders of Rwanda, and for the supply and sale to eligible customers in conformance with sector policies and other laws in force.

The Law also provides for a "Universal Access fund" to provide greater access to rural and other un-served areas.

2.2.2.2. Energy Policy

The Ministry of Infrastructure (MININFRA) developed a draft National Energy Policy whose principle objectives are to:

- a. Harmonize the National Energy Policy with Rwanda's long-term development plans and strategies;
- b. Give particular attention to requirements for the progressive development of the electricity sector to support economic development and the National Access Roll Out Program
- c. Have greater focus on household energy requirements and gender dimensions;
- d. Bring down the average cost of electricity supply
- e. Bring the (policy) statement up to- date by reflecting the latest and renewable energy and their environmental implications;
- f. State more clearly Rwanda's commitment to private sector participation and to regional cooperation in energy.
- g. clarify the roles and responsibilities of public sector agencies and develop public sector skills in planning, procurement, and transactions' negotiation
- h. Develop the legal, institutional and financial framework for rapid development of the electricity sector.

The Energy Policy is a comprehensive document, which addresses the principal issues in developing the energy sector in Rwanda. Some of the key issues include :

2.2.2.3 Energy pricing and subsidy policies:

Develop cost-reflective energy prices to ensure that energy suppliers can operate on a sustainable basis and make the necessary investments to expand power supply. Direct subsidies to one-time capital expenditures rather than to recurrent costs, and provide all subsidies in a transparent manner.

2.2.2.4. Regulatory framework:

Empower RURA and build its capacity to ensure independence in energy price regulation and licensing of energy providers.

Energy sector governance:

Management of the energy sector, including decision-making about projects, must be open and transparent and in the best interests of the country. Procurement should be rooted in the principles of transparency, equal treatment and non-discrimination between competing bidders.

2.2.2.5. Institutional framework and capacity building:

Strong energy sector institutions with adequate capacity are essential to meet ambitious growth targets. An effective energy information system is to be established and capacity building is to be provided to all sector institutions to undertake implement their roles and responsibilities.

2.2.2.6. Private sector participation in energy:

Private sector participation should be promoted at all segments of the energy supply industry. Where Public- Private Partnerships (PPPs) are desirable, government will work with private sector entities to ensure the speedy structuring and financing of PPP projects in the energy sector.

2.2.2.7. Financing energy sector investments:

GoR to leverage private sector financing with public financing, where appropriate. Reduce the need for government guarantees and contingent liabilities.

New and renewable energies:

Promote the use of renewable energy technologies that are financially, economically and socially beneficial.

Develop feed-in tariffs or other mechanisms to provide incentives and reduce risks for electricity production from renewable sources. Establish norms, codes of practice, guidelines and standards for new and renewable energy technologies.

2.2.2.8. Electricity regulation

Some of the key functions of Regulator, RURA, are to:

- i. conduct all technical regulatory activities for the power production transmission and distribution sectors
- ii. Issue permits and licenses to firms that satisfy licensing requirements
- iii. Monitor, evaluate and ensure the quality of the technical services provided by the electric utility
- iv. Ensure both compliance to the adopted standards and a fair competition between electricity operators
- v. Study and recommend tariffs and review and approve licensee tariffs
- vi. Promote sustainable provision of quality and safe services
- vii. Promote the utilization of renewable electrical energy resources in rural areas,
- viii. Promote energy efficiency and conservation measures.

The Electricity Law empowers RURA to set and approve electricity tariffs, in consultation with the Ministry and pursuant to laws and regulations in force. The Law also allows for cost based tariffs to ensure adequate return on investments made by license holders. The Law also allows for performance based pricing and benchmarking.

2.2.2.9. Electricity Tariffs

Rwanda has some of the highest electricity tariff in the region. The current electricity tariff is FRW 112/kWh (+VAT) for small Lv (low voltage) consumers, and FRW 105/kWh (+VAT) for large commercial and industrial Mv (medium voltage) consumers. A consultant study estimates that the tariff for residential and smaller non-residential customers is below the marginal cost of supply to residential customers, whereas the current industrial tariff is above the marginal cost of supply. The cost of supply is expected to reduce by 2012-13 when electricity production shifts from expensive diesel fuelled plants to cheaper hydropower and other generation options. The GoR has been supporting the power sector through:

- Direct operating cost support by paying for fuel imports/ equipment rental or exempting import-tax
- capital Expenditure support by seeking external funds as well as funds allocation from budget

- Other indirect subsidies feed-in tariffs (FIT) for eligible generation technologies is being considered and consultant studies are being reviewed to determine appropriate feed-in tariffs, especially for small hydro and other renewable energy sources. FIT for select generation technologies to be adopted by RURA by early 2012.

2.2.2.10. The Electricity Development Strategy

The objective of Rwanda's electricity strategy is to increase access to modern energy and to meet the ever increasing power demand for economic development of the country. To attain these objectives, the accelerated electricity generation mix proposed in the "Electricity Development Strategy 2011-2017", is to generate 1,000 MW from both the indigenous energy resources and from shared energy resources with neighboring countries. The following specific targets have been set in the Electricity Development Strategy:

- Hydropower generation to be increased to about 333 MW
- Geothermal power plants with capacity of 310 MW to be developed
- Methane gas to power projects will deliver 300 MW to the national grid
- 20 MW of additional diesel generation required for immediate power needs and serve as a back-up.
- 5 MW to be generated from renewable energy sources (solar Pv, micro hydro power or wind) and distributed to local communities beyond the national electricity grid
- Electricity connections to increase from 200,000 to a total of 1,200,000 by 2017, which will be equivalent to 70% of access
- Electrify 100% of schools, 100% of health facilities and 100% of sector offices by 2017, either through connection to the grid or through reliable off-grid systems
- Explore the possibility of developing all relevant projects as CDM projects right from the planning phase in order to sell emission reductions.
- Emphasize energy efficiency measures such as reduction of technical and commercial losses on the national grid, distribution of energy efficient lamps (CFL's) and the establishment of a Solar Water Heater subsidy scheme in order to decrease electricity costs and save energy (potential to save around 50 MW per year).

2.2.2.11. Power Market

EWSA is presently the sole off-taker for all power generated in Rwanda. Rwanda is also a member of the Eastern Africa Power Pool and plans to strengthen transmission interconnections with neighboring countries of Burundi, DR Congo, Tanzania and Uganda.

EWSA provides long-term power purchase agreement (PPA) to project developers.

Once the interconnection with neighboring countries is strengthened and the Eastern Africa Power Pool becomes operational, power can be exported through bilateral trades or to the power pool.

2.2.2.12. Demand Forecast

An Electricity Master Plan (EMP) has developed a demand forecast reflecting the goals of a new Electricity Strategy for the country, which envisages the development of 1,000 MW of generation capacity by 2017.

According to the EWSA data²¹, in 2009 its customers consumed approximately 307 million power (KWh) electricity which is 33% more than 231 million power (KWh) electricity in 2007. In 2009 the peak demand increased from 50.39MW to 63.26MW (21.6% increase). In June 2011, the peak demand was estimated to 97MW (41% growth compared to December data 2009).

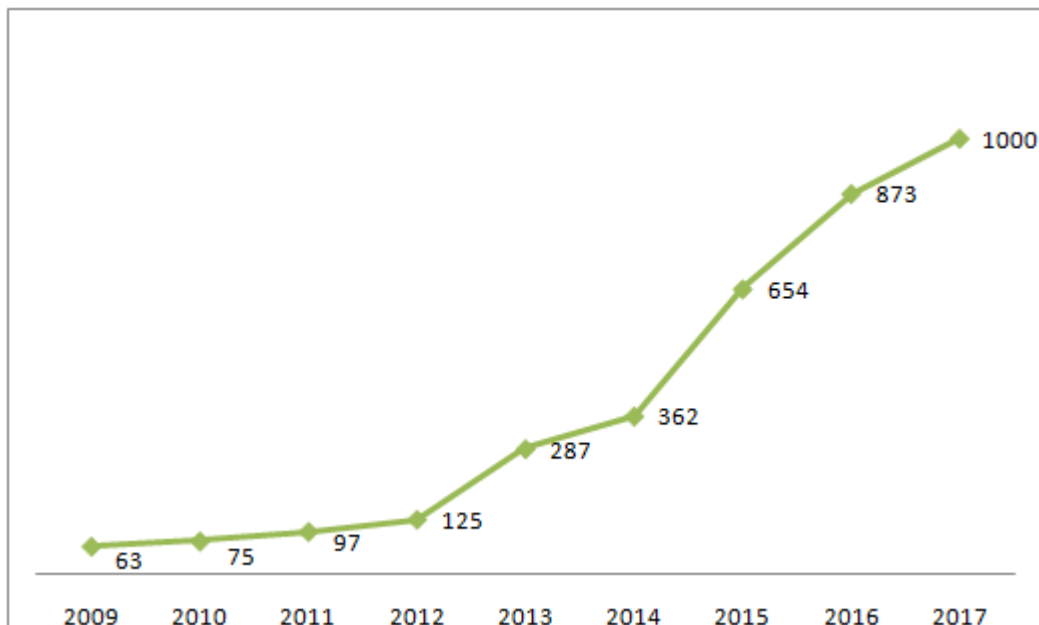


Figure 5: Electricity capacity demand forecasts for 2019-2017, in MW

2.2.2.13. Transmission network in Rwanda

Rwanda has about 383.6 km of 70kv and 110 KV high-voltage (HV) transmission lines, and about 4,900 km of and medium-voltage (Mv, 30 kv, 15 kv and 6.6 kv) lines and low-voltage (Lv, 380 v and 220 v) lines. Rwanda's electric network is interconnected with the networks of Burundi, the DRC and Uganda (there presently is no inter-linkage with Tanzania).

Power flows between Rwanda, Burundi and the Republic Democratic of Congo.

According to the Electricity Development Strategy for 2011-2017, Rwanda intends to extend its grid by 2,100 km (700 km of HV lines and 1,400 km of MV lines). In addition to 110 kV lines, 220 kV interconnection lines are planned to evacuate power from planned generation plants and meets the expected demand in the future construction of 400 kV lines is also under consideration within the framework of the interstate network development. Feasibility studies have been prepared, or are under preparation, for a number of transmission interlinkages including the 220 kV Kibuye- Kigali line,

The 220 kV:Kigoma–Rwegura (Burundi) and Birembo–Mbarara (Uganda) lines, and the 220 kV Rusomo– Kigali line.

2.2.2.14. Energy Sector Strategic Plan

During the EDPRS period, the main issues in the energy sector revolve around access to energy, costs of supply, energy security and the institutional framework in the management of energy. The specific issues that have been highlighted in the energy component of the EDPRS and therefore have been addressed by the Energy Sector Strategic Plan are to:

- i. Increase *access* to electricity for enterprises and households
- ii. Reduce the *costs of energy supply* while introducing cost-reflective tariffs
- iii. *Diversify* sources of energy supply and enhance energy security
- iv. Strengthen the governance framework and institutional capacity of the energy sector

The above can be met with the technical and social economic transformation of RE targeting small and medium-scale energy resources to increase the national generation capacity at an affordable RET.

The MDGs are an international initiative whose primary objective is to reduce global poverty. The initiatives identified 8 MDGs embracing economic, social and environmental dimensions of human development. Access to energy was not made one of the 8 MDGs, but analysis of the goals shows that energy services are an essential input into each of the primary MDGs.

Drawn from UNDP (2005): *Achieving the Millennium Development Goals: The role of Energy Services*

Table 3: *Energy and the MDGs*

MDG Target	Energy linkages for Economic Transformation
------------	---

1. Eradicate extreme poverty and hunger	Energy inputs such as electricity and fuels are essential for creating jobs, industrial activities, transportation, commerce, micro-enterprises and agriculture. To meet human nutritional needs, almost all staple foods must be cooked, which requires heat and fuels.
2. Achieve universal primary education	To attract teachers to rural areas, electricity generated from RE resources is needed for schools, and children need illumination after dusk to be able to study. Many children, especially girls, do not attend primary school as they must collect wood and water to meet family subsistence needs. Energy is also required to power ICT in education.
3. Promote gender equality and empower women	Adult women spend a large part of their day cooking and collecting water and fuel wood, which leaves them with little time for other productive activities. Without modern RE technologies and affordable stoves, and a lack of mechanical power for food processing and transportation, women often remain tied to drudgery.
4. Reduce child mortality	Diseases caused by lack of clean boiled water which can be easily got from RE resources in remote areas, and respiratory illness caused by the effects of indoor air pollution from traditional fuels and stoves, directly contribute to infant and child disease and mortality.
5. Improve maternal health	Lack of electricity in health clinics, poor illumination for night-time deliveries, and the daily drudgery and physical burden of fuel collection and transport, all contribute to poor maternal health conditions, especially in rural areas.
6. Combat HIV/AIDS, malaria and other diseases	Electricity is needed for radio and television, which can spread important public health information to combat deadly diseases. Health care facilities require electricity and the services that it provides (illumination, refrigeration, sterilization, etc.) to deliver safe, effective services.

7.Ensure environmental sustainability	Energy production, distribution and consumption can contribute to indoor air pollution, local particulates, land degradation, acid rain, and global warming. Cleaner energy systems are needed to address all of these issues to contribute to environmental sustainability.
8. Develop a global partnership for development	The World Summit for Sustainable Development (WSSD) called for partnerships between public entities, development agencies, civil society and the private sector to support sustainable development, including the delivery of affordable, reliable and environmentally sustainable energy services.

Source: UNDP/GTZ (2005): Scaling up Modern Energy Services in East Africa to alleviate poverty and meet the MDGs, East African Community.

The development impacts of energy-related interventions in the context of the MDGs have been analysed in a number of countries, with the following emerging as the key factors¹:

Considerable strides will have been made during the EDPRS (2008-2012) period in each of these dimensions. Beyond 2012, this experience will need to be built upon and extended initially to reach the MDG time horizon of 2015.

- i. Motive power – energy services that can be used for agricultural, manufacturing, transport and other livelihood activities – is a particularly important service for the poor.
- ii. Improvements in energy infrastructure – particularly electricity – are associated with industrialisation and reductions in poverty.
- iii. Energy services also play a critical role in improving education and gender equality.
- iv. Equally important is the impact energy services have on health.

2.3. Concept of Hydro Power plant.

A hydro scheme requires both water flow and a drop in height or 'Head' to produce useful power. The power conversion absorbs power in the form of head and discharge, and delivering power in the form of electricity or mechanical shaft power.

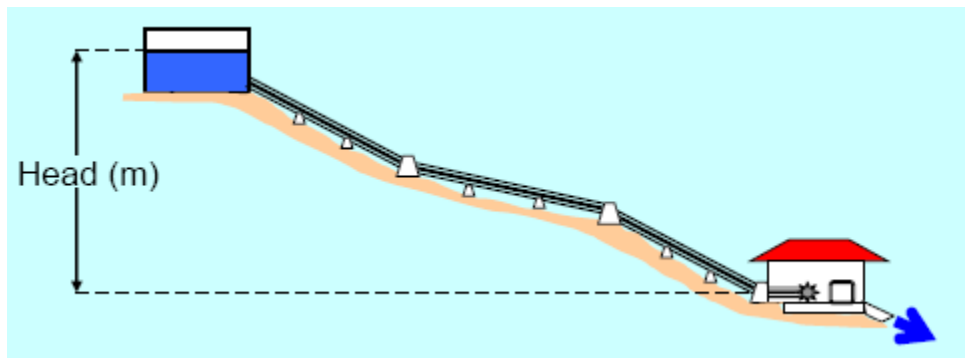


Figure 6. Power conversion in hydropower plant²⁵.

2.3.1. Micro-Hydro Power Plant Components

From the Figure 7, the components of micro hydro power plant are described below.

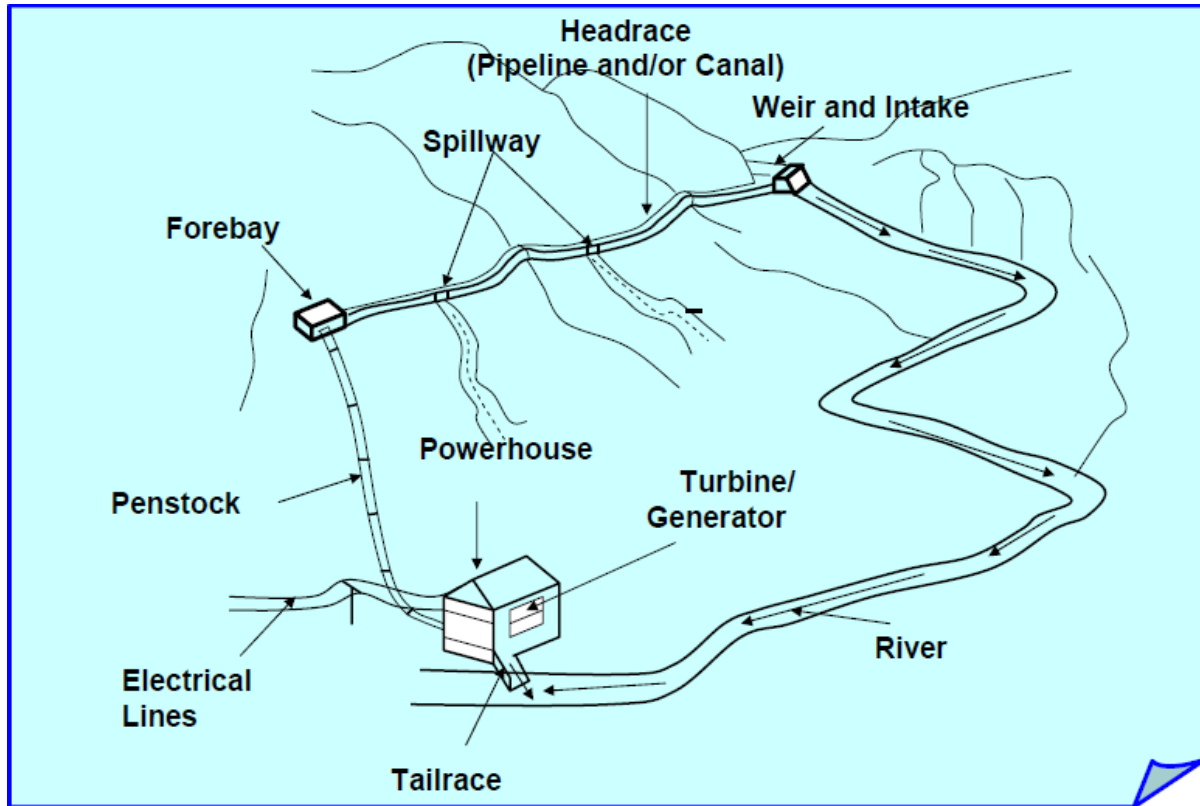


Figure 7 .Elements of micro hydro power plant (TEPCO, 2005)

2.3.2. Weir and intake

A hydro system must extract water from the river in a reliable and controllable way.

The water flowing in the channel must be regulated during high river flow and low flow conditions.

A weir can be used to raise the water level and ensure a constant supply to the intake. Sometimes it is possible to avoid building a weir by using natural features of the river. A permanent pool in the river may provide the same function as a weir.²⁶

The intake of a hydro scheme is designed to divert a certain part of the river flow.

This part can go up to 100 % as the total flow of the river is diverted via the hydro installation. For small systems only a tiny fraction of a river might be diverted, this also has the advantage

that Micro Hydro power (MHP) output can be kept constant even when the flow of the river is strongly fluctuating.

The following points are required for an intake²³:

- the desired flow must be diverted,
- the peak flow of the river must be able to pass the intake and weir without causing damage to them,
- as less as possible maintenance and repairs,
- it must prevent large quantities of loose material from entering the channel,
- it must have the possibility to remove piled up sediment.

Different types of intakes are characterized by the method used to divert the water into the intake. For micro hydro schemes only the small intakes will be necessary. The main type of intake for such purposes will be the side intake since it is cheap and simple to construct.

As no flow data is available, it is possible to use hydrological methods that are based on long-term rainfall and evaporation records, and on discharge records for similar catchment areas. This allows initial conclusions to be drawn on the overall hydraulic potential without taking actual site observations. But, these data are not available at all, and then it is advisable to follow this up with site measurements once the project looks likely to be feasible and the assumptions are made for this case according to the site observations.

2.3.3. Headrace, Forebay

Headrace is the channel which conducts the water from the intake to the forebay tank.

The length of the channel depends on local conditions. In one case, a long channel combined with a short penstock can be cheaper or necessary, while in other cases a combination of short channel with long penstock suits better.

Most channels are excavated, while sometimes structures like aqueducts are necessary. To reduce friction and prevent leakages channels are often sealed with cement, clay or polythene sheet.

Size and shape of a channel are often a compromise between costs and reduced head. As water flows in the channel, it loses energy in the process of sliding past the walls and bed material. The rougher the material, the greater the friction loss and the higher the head drop needed between channel entry and exit.

Incorporated in the channel are the following elements: settling basin (removes sediments from water), spillways (used for controlled overflow) and forebay tank.

The forebay tank forms the connection between the channel and the penstock. The main purpose is to allow the last particles to settle down before the water enters the penstock. Depending on its size it can also serve as a reservoir to store water.

2.3.4. Penstock

Penstock is a covered pipe which is used to convey water from the Forebay tank to the turbine inlet by keeping the pressure inside. This constitutes a major expense of a micro hydro budget. Hence it is wise to optimize the penstock design considering the following²³.

- Penstock size and its thickness
- Material of penstock
- Selecting the terrain
- No. of supports, size and their stability.
- No. of bends, anchor blocks and their stability.

In many cases mild steel and High Density Poly Ethylene (HDPE) pipes proved to be the most economic solution for penstocks for micro hydro schemes. However, many aspects, like availability, costs, weight, stability etc., have to be taken into account for any specific site. The penstock alignment should be chosen such that significant head can be gained at a short distance but still be possible to lay the penstock and build support and anchor blocks on the ground. The number of bends on the alignment should be kept to a minimum so that the number of anchor blocks and head loss can be minimized.

We note that in case of HDPE, pipes are used as penstock; these should be buried to a minimum depth of 1 m. Similarly, if mild steel penstock pipe needs to be buried a 1 m burial depth should be maintained and corrosion protection measures such as high quality bituminous paints should be applied. Due to higher risks of leakage, flange connected penstocks should not be placed underground. For the safety of penstock line and reliability of the hydro projects, it is essential to allocate for expansion joints, couplings, reducers, bends and vent pipes where necessary.

Chapter3. Hydrology and Geology

3.1. Hydrology

3.1.1. Introduction to hydrology

The main objectives of the hydrologic study of a small or micro hydropower scheme are the characterization of:

- The run off at the water intake of the scheme in order to allow the determination of the design discharge, and, thus, the design of the water intake, of the diversion circuit and of the powerhouse, as well as the evaluation of the energy production.
- The floods or, more precisely, the peak flows, to consider in the design of the weir, of some of the diversion works and of the powerhouse (for instance, if the turbines are of the Francis type they should be located above the water surface elevation in flood conditions, at the powerhouse outlet)²⁶

The amount of energy that can be generated depends on the amount of water available in a river. The determination of the amount of water available in the river and its distribution throughout the year is vital at the planning and design stages of a hydropower scheme¹⁷ So, a long record of discharge in the river is necessary, though, such discharge records for long periods are not often available¹⁷ However, long records of rainfalls are available and based on knowledge in hydrology as such records can be used to estimate discharges in rivers. Using the discharge records, the availability of water can be determined based on flow-duration relationships.

Maximum floods that can be expected in a river are also very much required to design various hydraulic structures in a hydropower project ¹⁷.

3.1.2. Basic information required for the hydrologic study

The basic hydrologic data required for the evaluation of the energy production in a small hydropower scheme is the mean daily flow series at the scheme water intake in a period that has to be long enough in order to represent, in average, the natural flow regime ²⁶. Therefore, it is reasonable to assume that the errors of the estimates that result from the variability of the natural flows are minimized ²⁶.

All the relevant data were collected from the relevant authorities of the government. The following data required for the study were collected.

- Maps of the catchment area,
- Rainfall data,
- Temperature,

The rainfall data and the temperature data are essential to calculate the hydrological responses of the catchments. The accuracy at which the analysis is done should be adequate for proper estimation of the diverted water from the given rivers ¹⁷.

Time averaged data considering small periods for large number of such periods can provide the flow duration curves with same accuracy as the detailed analysis shows ¹⁷.

Therefore, in this study the data were collected in monthly intervals but for long series as availability permits.

3.1.3. Rainfall

There is one nearest rain gauging station available within the catchment. The details of the gauging station are given below.

Table 3. Location of the influencing rainfall gauging station

	Latitude	Longitude	Period of data availability
Muganza	-1.72 ⁰	29.85 ⁰	1970-1993

There were a few missing data in the rainfall records and they were filled by using the rainfalls in the other years at the same station.

Data for 23 years from 1970 to 1993 were used in the study. Table 4 shows the mean of monthly rainfalls of the Muganza station used in the analysis. Figure 22 presents monthly rainfall distribution of the station ³².

Intra-annual variability of total monthly rainfall and mean monthly temperature from selected climate stations, illustrated for the data of 1970–1993, is presented in Figure 23 and Figure 24

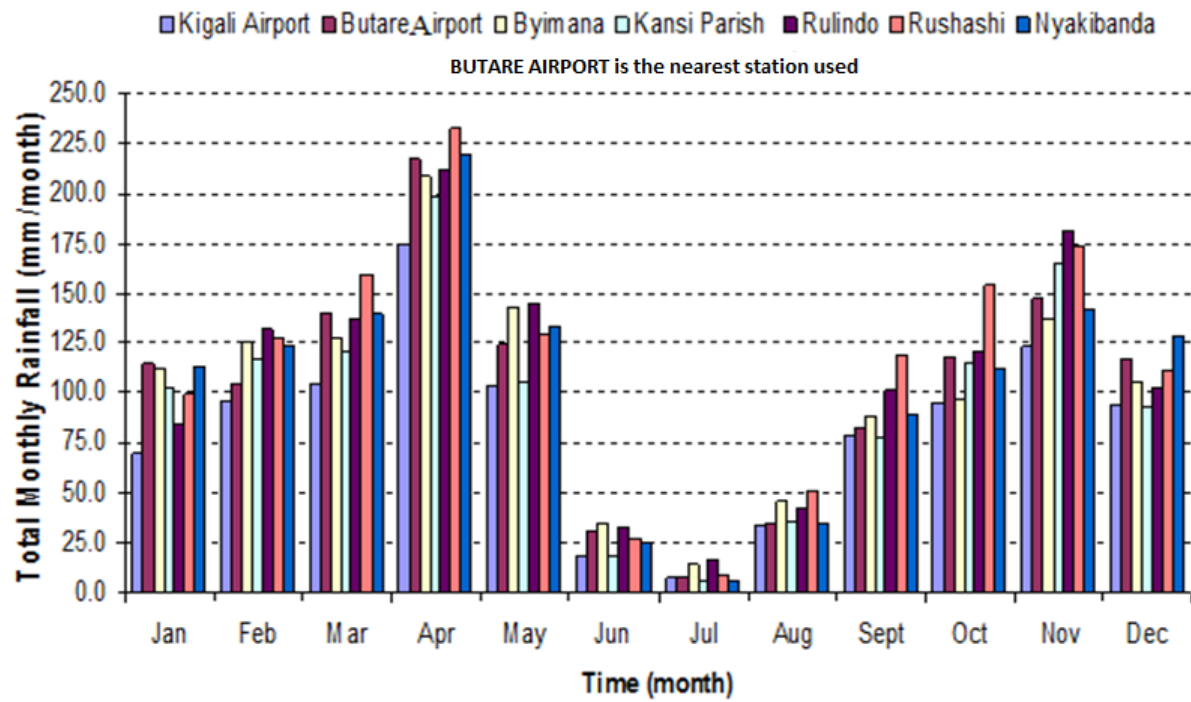


Figure 8. Intra-annual variability of total monthly rainfall, illustrated for the data of 1970–1993 (Omar, 2010).

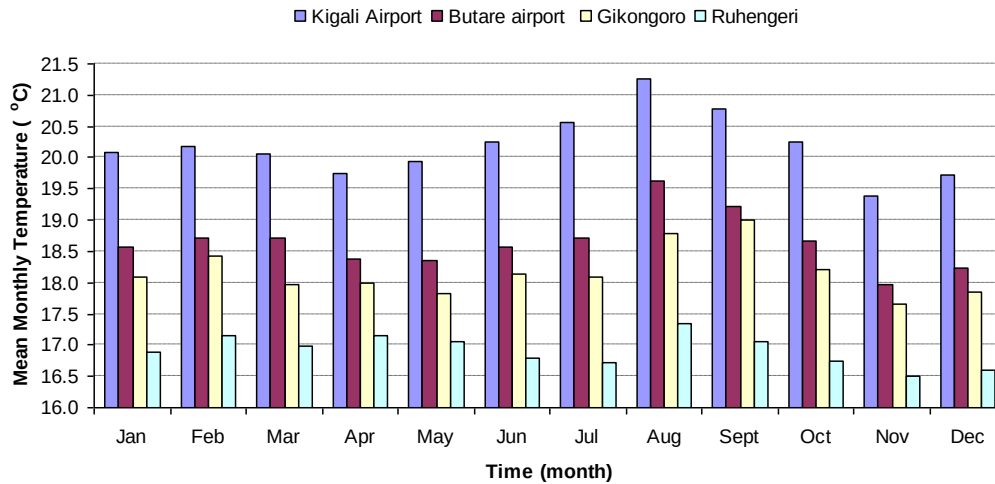


Figure 9: Intra-annual variability of total mean monthly temperature, illustrated for the data of 1970–1993 (Omar, 2010).

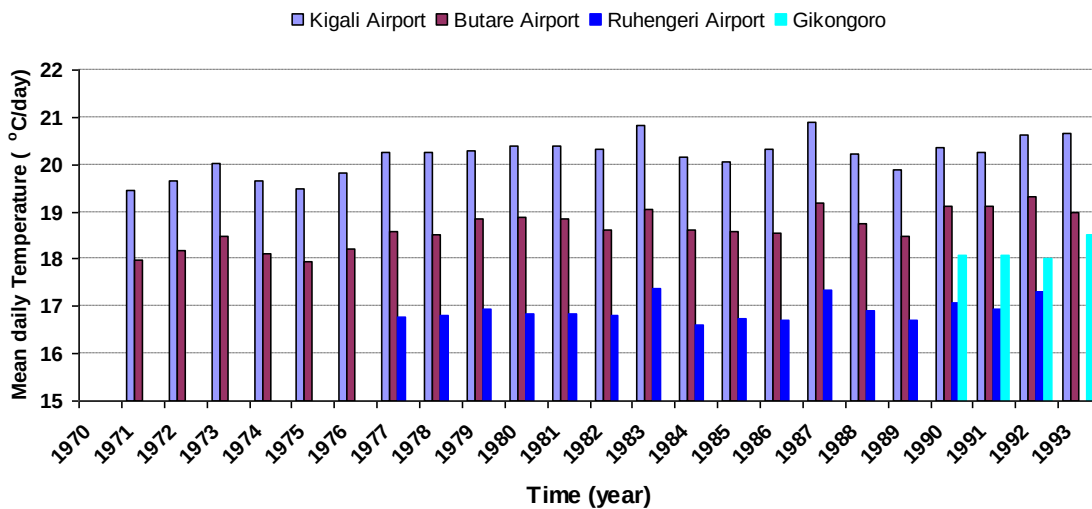


Figure10: Inter-annual variability of temperature (Omar, 2010) (Data used GISUMA MHP).

From the inter-annual variability of temperature, based on selected stations of Rwandan catchments (Kigali, Butare, Ruhengeri and Gikongoro); data covers 1970-1993 we can make the following assumption;

The data from Butare airport station can be used as the closest to GISUMA catchment area from the year 1970 to 1993. The mean minimum daily temperature in the whole time series at Butare airport station is approximately the same to the temperature values compare to GISUMA catchment area due to the same topography of these two different catchments. The altitude of Butare airport station is 1493m whereas the downstream altitude for GISUMA catchment area is 1400 m.

Figure10 shows the record of daily mean temperature data for 1970-1993 with mean yearly temperature of 18.6⁰C. For the selected time series data records at Butare Airport.

From the graph of the Figure9 we can complete the table to consider Muganza as nearest rainfall station of Gisuma catchment area and draw the monthly average rain gauge values at Muganza as follows:

Table 4. Monthly average rain gauging station at Muganza

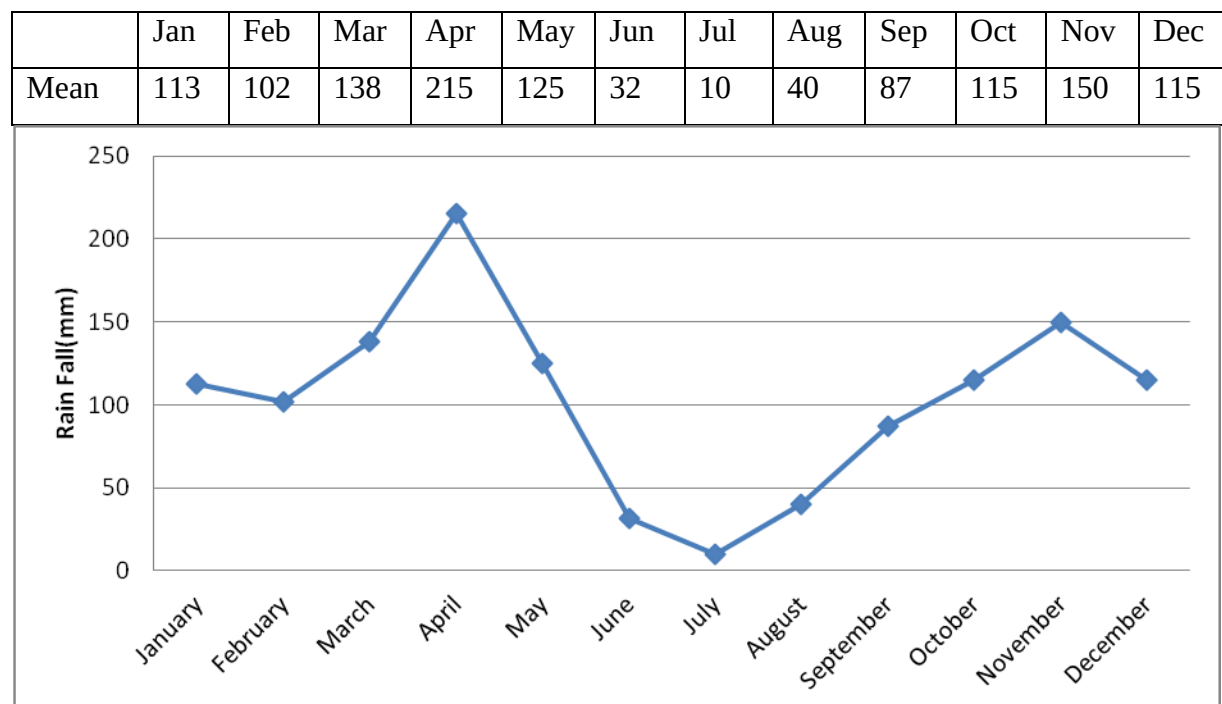


Figure 11. Mean rainfalls at the rainfall gauging station Muganza

Figure 4.1 shows the annual rainfall variation over the 23 years period considered in the study .It also shows the rainfall trend, which is found to be slightly increasing²³ .The interesting behavior is that the dry year rainfalls as well as the wet year rainfalls are increasing. This yields better prospect for run-of-the river type micro-hydropower schemes because this means that the low flows will increase with time.

The annual average flow variation in the river is given in the Figure 12 As the figure depicts; there is a very small decreasing trend in the annual average flow in the Gisuma river⁵

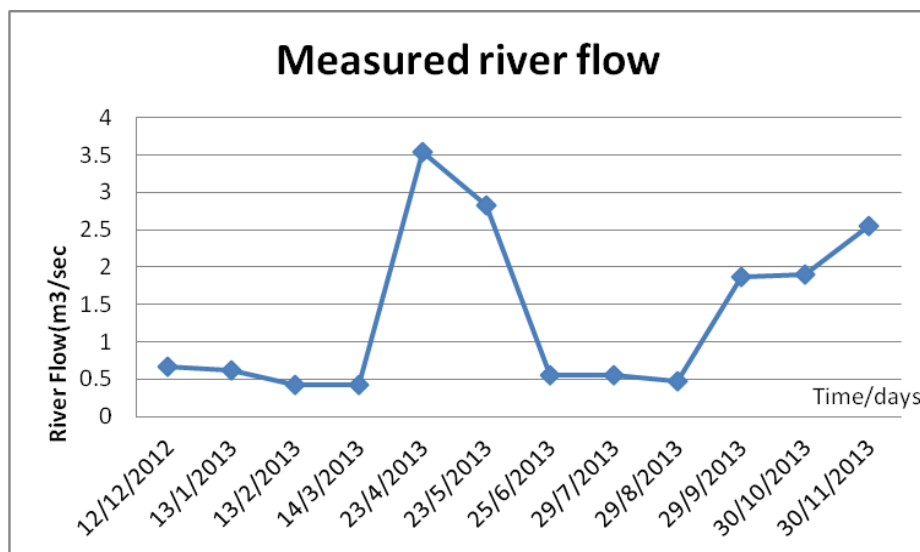


Figure12: Annual average flow variation of Gisuma river.

3.1.4. Flow Duration Curve

Based on the calculated stream flows (Table 5), the flow duration relationship was determined for the whole analysis period. Figure 28 shows the developed Flow- Duration Curve for the river¹⁰. As it shows a flow of about 1.3 m³/s is available in the river for more than 80% of the time. However, for more than 50% of the time a flow of 1.5 m³/s is available in the river.

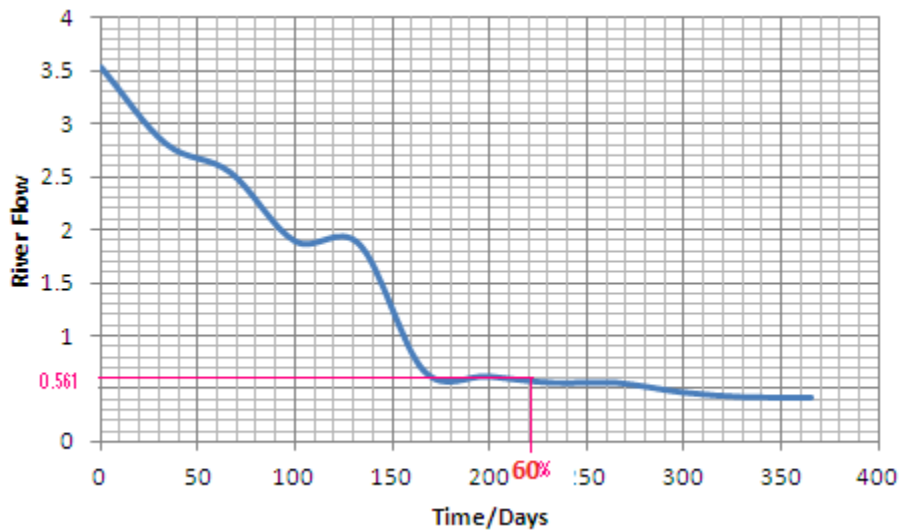


Figure 13 : Flow duration curve for the considered period.

3.1.5. River Gauging

The river discharge (stream flow) is the volume of water in the river that flows through a point in a certain time. The measurement, or gauging, of river discharge is important because river discharge values are necessary for the calculation of surface water resources. These values give information about the maximum and minimum volumes of water flowing in the river, which is required for planning and designing hydroelectric projects. Among the stream flow measurement techniques, velocity-area method is the mostly used method.

The velocity-area method is based on the continuity Equation.

Discharge is determined by measuring cross sectional area and the velocity. The cross-sectional area of a river channel at some point is determined from measurements of the depth of the water taken at known intervals across the river. The width is subdivided into a number of subsections depending upon the degree of variability of the depth across the stream and the degree of precision required.

3.1.6. Hydrological model

To calculate run-off, the effective rainfall has to be estimated. The following steps were adopted in the estimation of effective rainfall. Initially the potential evapotranspiration were estimated based on the Blaney Criddle formula. The Blaney Criddle equation is a relatively simplistic method for calculating evapotranspiration for periods of one month or greater. When sufficient meteorological data is available the Penman–Monteith equation is usually preferred. The Blaney Criddle equation is however ideal when only air temperature data is available for a site

$$ET = kp (0.46T_a + 8.13) \quad \text{Equation (4.2)}$$

Where: ET potential evapotranspiration from a reference crop, in mm, for the period in which p is expressed;

$$T_a \text{ is the mean daily temperature } [^{\circ}\text{C}] \text{ given as } T_a = (T_{\max} + T_{\min}) / 2 \quad \text{Equation (4.3)}$$

P = percentage of total daytime hours for the used period (daily or monthly) out of total daytime hours of the year (365×12);

k = monthly consumptive use coefficient, depending on vegetation type, location and season and for the growing season (May to October), k varies from 0.5 for orange tree to 1.2 for dense natural vegetation.

Following the recommendation of Blaney and Criddle, in the first stage of the comparative study, values of 0.85 and 0.45 were used for the growing season (April to September) and the non-growing season (October to March), respectively.

An infiltration loss of 1% of the rainfall is assumed in the estimation of effective rainfall. The effective rainfall is the total rainfall less the losses due to actual evapotranspiration and infiltration. Subsequently, using the effective rainfall, the stream flows were calculated based on the following relationship.

$$\text{Flow} = \text{Effective rainfall} \times \text{catchment area}$$

Evapotranspiration is the sum of evaporation and plant transpiration from the Earth's land surface to atmosphere. Evaporation accounts for the movement of water to the air from sources such as the soil. Evapotranspiration is an important part of the water cycle.

Potential evapotranspiration or PE is a measure of the ability of the atmosphere to remove water from the surface through the processes of evaporation and transpiration assuming no control on water supply. Actual evapotranspiration or AE is the quantity of water that is actually removed from a surface due to the processes of evaporation and transpiration.

3.2. Geology of the Area

This section presents methodology of the investigation, geological and geotechnical aspects of the proposed site.

3.2.1. Methodology

The project core activities involve the design, costing and feasibility study of the use of micro hydropower plant for improving the accessibility of renewable energy situation in rural areas of Rwanda.

The methodology employed to undertake the study includes: literature search and review, description of equipment used to undertake the survey work on the site to determine the flow rate of the river and other important parameters. Brief discussions and interview with local people during community meetings that were organized during the site visit, sources of socio-economic data and the information collected during the site visit can be used to design, cost and determine financial viability and feasibility of the Gisuma Micro-Hydro power plant.

3.2.2. Methods of data Collection.

- i. Reconnaissance and field observation on the site,
- ii. Literature and official documents study for some previous recorded data from different institutions dealing with environmental and natural resources like Ministry of Infrastructure (MININFRA), Ministry of Natural Resources (MINIRENA),

RWANDA Metrology Center,

- iii. Available head, minimum and maximum flow rates, rainfall data, catchment area considered for run-off, number of houses in the villages to be supplied by Gisuma micro hydro power plant, etc
- iv. Data analysis,
- v. Design and costing of the Gisuma micro hydro power plant.

The appropriate and recommended tools for the field investigations cover the following issues.

- ✓ General and site information,
- ✓ Technical specifications (including available run-off, water usage, etc.),
- ✓ Domestic and public electricity demand,
- ✓ Commercial electricity demand,
- ✓ To estimate income from energy supply,
- ✓ Environmental issues,
- ✓ Estimated project costs.
- ✓ Estimated project payback period using present electricity tariffs.

Initial site investigations were carried out during December, 2012 for the first time and July, 2013 for the second time, followed by calculations using the data got from different instruments used during site visit etc prior to the compilation of final study report.

3.2.3. Site locating

Locating the sites for the weir, path of the channel and penstock and the powerhouse properly is very important for the sustainability and optimum performance of any hydropower generation scheme. The weir should be located in such a way to utilize the maximum possible and available energy head. At the same time the location is relevance to the river morphology should minimize the sediment intrusion to the diversion and should support optimum diversion facility for the required quantity. The space available in the river at the barrage location should be enough to locate all the necessary structures for control of the diversion and intake.

In case where the river morphology is unfavorable for the control of sediment intrusion proper design measures should be adopted to avoid or minimize the sediment intrusion. The location and the river geometry also should allow economizing of the structural design and should favor stable design both structurally and hydraulically. The channel and penstock also should favor structurally stable and economical design as well as easy construction because generally entire channel traces are not easily accessible. Stable mountain slopes, erosion control of the slopes are generally required conditions to meet stability of the channels.

The powerhouse should be located to enhance the full utilization of the available energy head. At the same time the location should support a structurally sound and economical design of the powerhouse and the machine floor. As the powerhouse is located close to the river to enable efficient discharge of the used water to the river it should be situated above the high flood level.

This section presents the details of various alternative configurations considered during the reconnaissance survey at the proposed site of the GISUMA micro-hydropower plant. Out of the numerous alternatives the major alternatives that may be feasible and worth considering are discussed here.

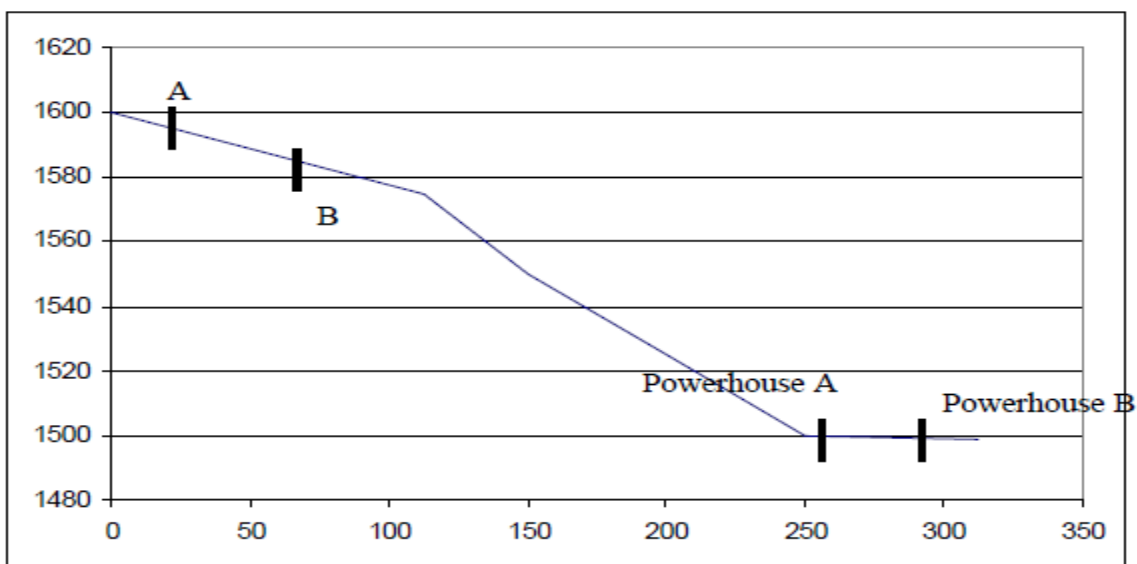


Figure 14 : Longitudinal section of the river

The Figure 14, shows the longitudinal section along the river starting close to the proposed barrage site up to the location of the powerhouse. Looking at the above Figure and the site conditions the decision on locating the barrage was reached at the preliminary site reconnaissance survey. Two barrage locations are suggested at this initial reconnaissance survey. They are shown as locations 'A' and 'B' on the Figure 14. Any other locations downstream of the location B would not be feasible due to be located at nearest of irrigation land to be cultivated. Any other location upstream of the location A would not be feasible as it corresponds to the shift upstream. The Locations A and B are located on either side of small hill rock around which the GISUMA river flows. Both these locations try to maintain the same barrage height.

3.2.4. Measuring weirs¹⁸

A flow measurement weir what we will use has a rectangular notch in it through which all the water in the stream flows. It is useful typically for flows in the region of 50-1000 l/s. The flow rate can be determined from a single reading of the difference in height between the upstream water level and the bottom of the notch Figure15. For reliable results, the crest of the weir must be kept 'sharp' and sediment must be prevented from accumulating behind the weir

The formula for a rectangular notched weir is:

$$Q = \frac{2}{3} C_d \sqrt{2g} (L - 0.2h) h^{1.5} \quad \text{Equation (2.1)}$$

Where:

Q = flow rate (m^3/s)

C_d = the coefficient of discharge

L = the notch width (m)

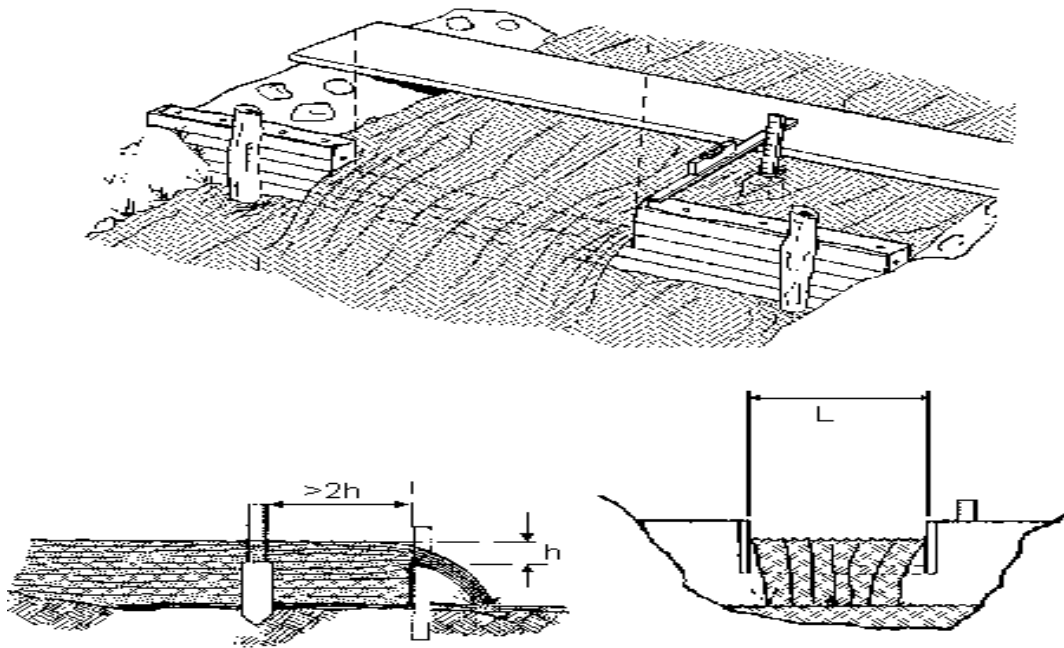


Figure 15 : Measuring Weir method

h = the head difference (m)

g = acceleration due to gravity (9.81m/s^2)

If C_d is taken, typically, as 0.6, then the equation becomes:

$$Q = 1.8 (L - 0.2h) h^{1.5} \quad \text{Equation (2.2)}$$

Since stream flow varies both from day to day and with the season, measurements should ideally be taken over a long period of time, preferably several years.

3.2.5. Measurement of Head

The head of water available at any one site can be determined by measuring the height difference between the water surface at the proposed intake and the river level at the point where the water will be returned.

The use of a Dumpy level (Theodolite or builder's level) is the conventional method for measuring head accurately and should be used wherever time and funds allow. Such equipment should be used by experienced operators who are capable of checking the calibration of the device.

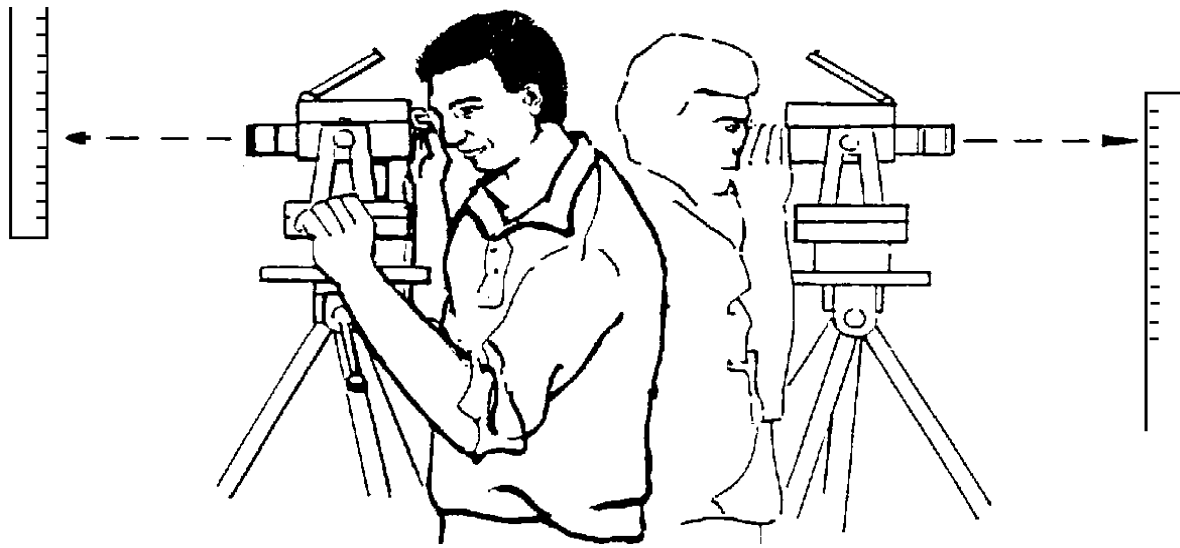


Figure 16: Measuring of head using Theodolite

A measurement of the available head (H) at a site is required to complete an accurate calculation of power output. When determining head, both gross head and net head must be considered.

“Gross head (Figure17) is the vertical distance between the top of the penstock that conveys the water under pressure and where the water discharges from the turbine. Net head is the available head after subtracting the head loss due to friction in the penstock from the gross head”

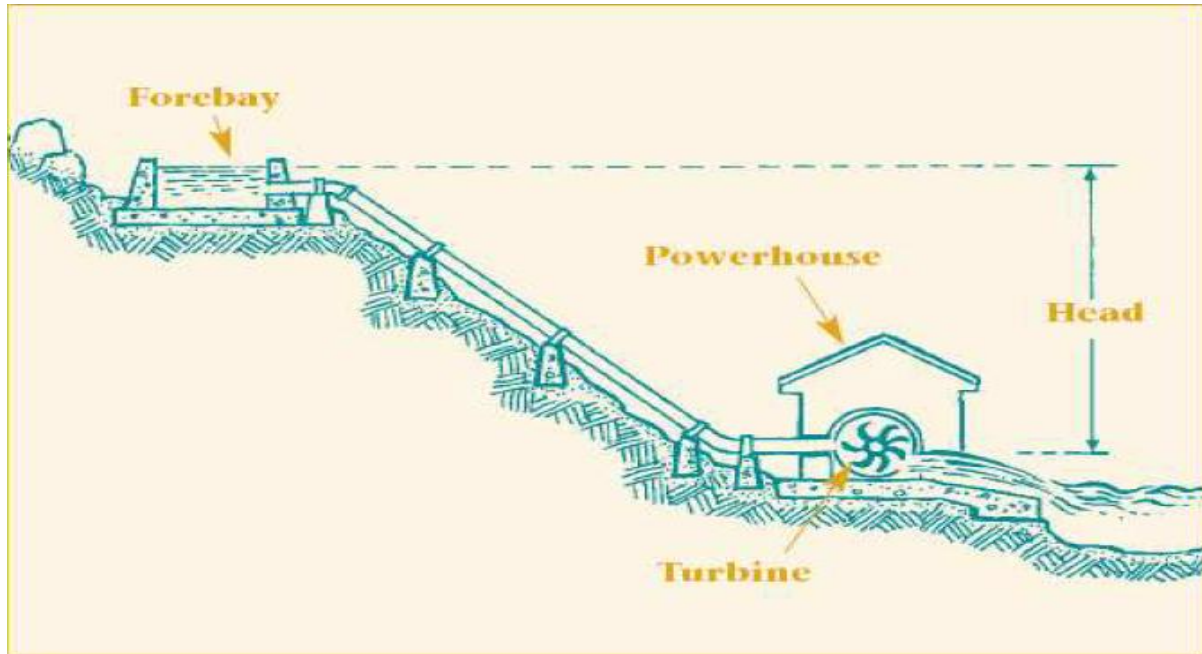


Figure 17 : Head Measurement of a Micro-Hydropower System

Gross Head

There are methods of assessing gross head applicable to low-head sites and others more suited to high-head sites. Field measurements of gross head are usually carried out using surveying techniques. The precision required in the measurement will limit the methods that can be employed. In the past, the best way to measure gross head was by levelling with a surveyor's level and staff; however this was a slow process.

Accurate measurements were made by a tachometer or less accurately by a clinometer or Abney level. Nowadays with digital theodolites, electronic digital and laser levels and especially with the electronic total stations the job has been simplified.

Gross head of the project is measured as 131.5 m from the site surveying. The net head is calculated on the conceptual design chapter.

3.2.6. Measurement of Flow Rate

Direct measurement of flow rate is not possible, but must be calculated from velocity and cross-sectional area of the stream.

One common method used for velocity measurement of medium to large rivers involves the use of mechanical current meter or float method, etc., and the measurement of the cross sectional area of the river involves the use of mid-section method as explained below. Ideally the part of the river analyzed should be a straight run with little water turbulence. The “stage” (the top water level) of the river is measured by analysis of the varying water levels in the waterway over a period of time (preferably a year) by use of a staff fitted into the water.

Water levels are noted at the same time every day. A rating curve is then generated by the correlation of these daily “stage” measurements and the discharge being defined as “the volume per unit time that passes any point in a stream”.

Once the velocity (v) and cross-sectional area (A) of the stream are obtained, the calculation of discharge (Q) would be: $Q = V.A$

Velocity measurement methods:

➤ By float

A floating object, which is largely submerged (for instance a wood plug or a partially filled bottle), is located in the centre of the stream flow. The time t (seconds) elapsed to traverse a certain length L (m) is recorded. The surface speed would be the quotient of the length L and the time t . To estimate the mean velocity, the above value must be multiplied by a correction factor that may vary between 0.60 and 0.85 depending on the watercourse depth and their bottom and riverbank roughness (0.65 is a well accepted value). The accuracy of this method is dependent on the range of correction factor.

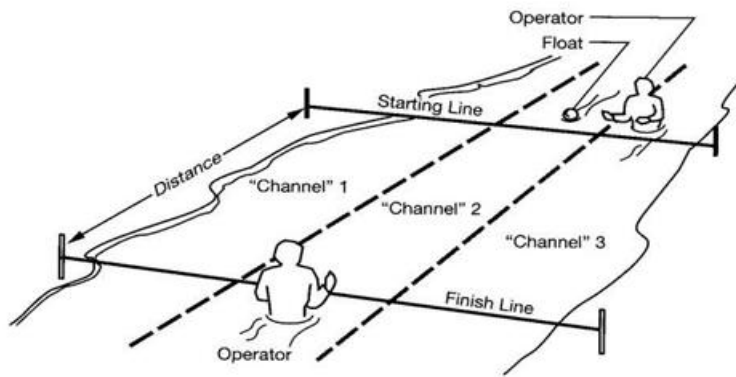


Figure 18 Float Method of Measurement (Padden, 2010)

➤ By mechanical current-meter

A current-meter is a fluid-velocity-measuring instrument. Current meters are classified into two types: - Vertical axis rotor with cups: This type of instrument has a circle of small conical cups, disposed horizontally which rotate about the suspension axis. These current meters operate in lower velocities than the horizontal axis rotor types, and have the advantage of bearings being well protected from silty waters.



Figure 19: Mechanical Water Current Meter .

Horizontal axis rotor with vanes (propeller): A small propeller rotates about a horizontal shaft, which is kept parallel to the stream by tail fins. The instrument is weighted to keep it as directly

as below the observer as possible. This rotor has the advantage of being less likely to disturb the flow around the measuring point and also for being less likely to become entangled by debris.

Each revolution of the propeller is recorded electrically through a cable to the observer and the number of revolutions is counted by the observer, or automatically by the instrument itself, over a short period (say 1 or 2 minutes). These observations are converted into water velocities from a calibration curve for the instrument. By moving the meter vertically and horizontally to a series of positions (whose coordinates in the cross-section are determined), a complete velocity map of the cross-section can be drawn and the discharge through it calculated.

Discharge measurement, or stream gauging, requires special equipment and, sometimes, special installations. The measurements should be made by an agency that has staff with expertise in the techniques of hydrological survey. The most accurate method is to measure the cross-sectional area of the stream and then, using a current meter, determine the average velocity in the cross-section.

Velocity varies approximately as a parabola from zero at the channel bottom to a maximum near the surface. A typical vertical velocity profile is shown in Figure 20. It has been determined empirically that for most channels the velocity at six-tenths of the total depth below the surface is a close approximation to the mean velocity at that vertical line. However, the average of the velocities at two-tenths and eight-tenths depth below the surface on the same vertical line provides a more accurate value of mean velocity at that vertical line.



Figure 2 Typical river velocity profiles in the vertical plane

Since the velocity both across the flow and vertically through it is not constant, it is necessary to measure the water velocity at a number of points to obtain a mean value. Velocity also varies across a channel, and measurements must, therefore, be made at several points across the channel. The depth of the river varies across its width, so the usual practice is to divide the cross-section of the stream into a number of vertical sections as shown in Figure 21 and measure velocity at each of these.

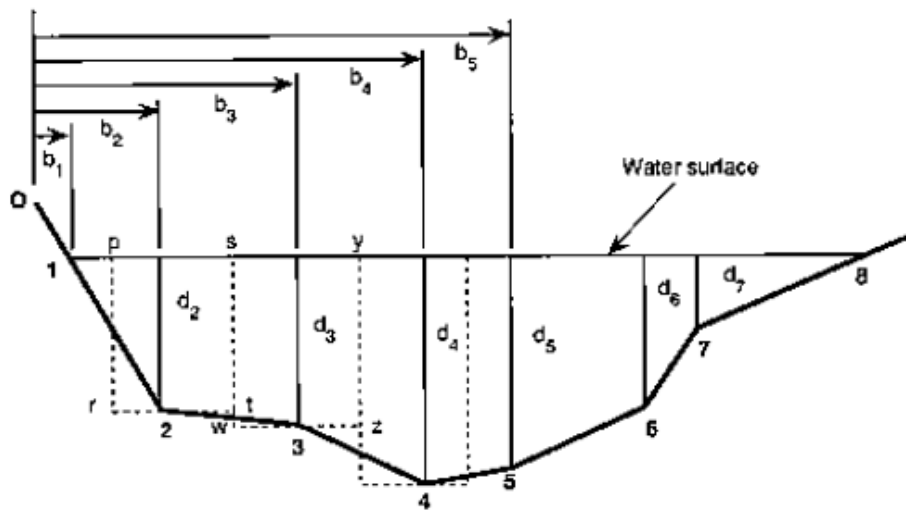


Figure 21 Cross-section of a stream divided into vertical sections for measurement of discharge

Procedure for measuring discharge¹

- All measurements of distance should be made to the nearest centimeter;
- Measure the horizontal distance b_1 , from reference point 0 on shore to the point where the water meets the shore, point 1 in Figure 21;
- Measure the horizontal distance b_2 from reference point 0 to vertical line 2.
- Measure the channel depth d_2 at vertical line 2.
- With the current meter make the measurements necessary to determine the mean velocity v_2 at vertical line 2. Repeat steps 3, 4 and 5 at all the vertical lines across the width of the stream.

The computation for discharge is based on the assumption that the average velocity measured at a vertical line is valid for a rectangle that extends half of the distance to the verticals on each side

of it, as well as throughout the depth at the vertical. Thus, in Figure 21, the mean velocity V_2 would apply to a rectangle bounded by the dashed line p, r, s, and t.

The area of this rectangle is

$$a_2 = \frac{b_3 - b_1}{2} \times d_2 \quad \text{Equation (2.3)}$$

and the discharge through it will be:

$$Q_2 = a_2 \times V_2 \quad \text{Equation (2.4)}$$

Similarly, the velocity applies to the rectangle s, w, z, y and the discharge through it will be:

The discharge across the whole cross-section will be:

$$Q_T = Q_1 + Q_2 + Q_3 \dots Q(n-1) + Q_n$$

In the example of Figure 12, $n = 8$. The discharges in the small triangles at each end of the cross-section, Q_1 and Q_n , will be zero since the depths at points 1 and 8 are zero.

Chapter4. Environmental impact assessment and its identification

The purpose of an environment impact assessment applied to micro-hydro power plant consists in the evaluation of the favorable and unfavorable impacts, in what concerns the two categories of impacts, in natural and social environmental context. Natural impacts include hydrology and sediment effects, as well as the water temperature and quality, ecology, engineering construction, biology, soils and geology, noise and, eventually local, climate change. Social impacts involve social, cultural and economic development inducing local industrialization and changes in citizens' quality of life as well as potential people displacement due to submersion by the reservoir.

➤ 4.1 Water quality

Water of Gisuma river as a natural Water river is never completely pure, is varying, considerably, in the range and in concentrations of dissolved substances present, such as suspended particles, pH and temperature. Fast-flowing streams and rivers, normally, have high oxygen levels absorbed from the atmosphere due to natural turbulence that why it needs to be treated carefully before using it. Nevertheless, in the Gisuma river, there is no significant change in water temperature. During the site visit some of the said parameters are measured.

➤ **4.2 Air and pollution**

Gisuma hydropower plant like other micro- hydropower plants produce no carbon dioxide, sulphur oxides or nitrous oxides, no air emissions and no solid or liquid wastes as their machinery does not use pollutant fuels which generates harmful fumes. Nevertheless there are impacts by retaining water and inducing sediments to settle down.

The worlds concern about planet global warming phenomenon essentially due to CO₂, SO₂, NO_x emissions in energy generating process with fossil fuels and the problem for the future of nuclear wastes, more and more will be emphasized in the advantages of energy production through renewable sources. Micro-hydropower represents an important environmental benefit to aid sustainable development because there is no release of carbon dioxide that contributes to ozone depletion and global warming.²⁹

Gisuma Micro-hydropower does not require high dam because it is a run-of-river scheme, meaning simply that the turbine only generates when there is available water. A minimum daily storage and flow regulation is typically guaranteed. When the river dries up the generation ceases.

Other general Gisuma Environmental Issues

The most central environmental problems and issues and possible remedial measures are:

- Loss of land, scenic beauty and other natural resources.
- Relocation of people, homes, economic activities and infrastructure.

- Removal of trees and other biomass from area to be flooded.
- Use of reservoir for, recreation, water supply, irrigation, etc,
- Reservoir water body as source of water borne diseases.
- Loss of water through evapotranspiration

These factors are diverse and are project specific. Each issue and its relationship to other factors needs to be analyzed individually and specifically in its real setting: general conclusions on mitigative actions will have little meaning.

Separate development project will be identified where relocation of significant number of people is involved. The preferred solution will be to work with the affected people and create a demand for relocation among them and thereby avoid forced displacement. Further to this the following environmental issues may arise in the downstream.

Chapter5. Conceptual Designs

5.1. Measurement of the discharge, pH, Temperature¹⁹

In order to calculate the flow of GISUMA River, the velocity and cross-sectional area was determined before, as the depth of the river varies, the best method used was to divide the stream into sections and measure the depth of each section. The cross-sectional area was then calculated for each section, and the sum of the areas of all the sections was been the cross-sectional area of the river. The velocity is then measured using a floating method, which is shown in Figure 22, Figure 23 and Figure 24 shows the measurement of electrical conductivity and temperature.



Figure 22 : Measuring velocity using floating method.



Figure23: Velocity of water flow measurement at the site using a floating object and a stop watch.

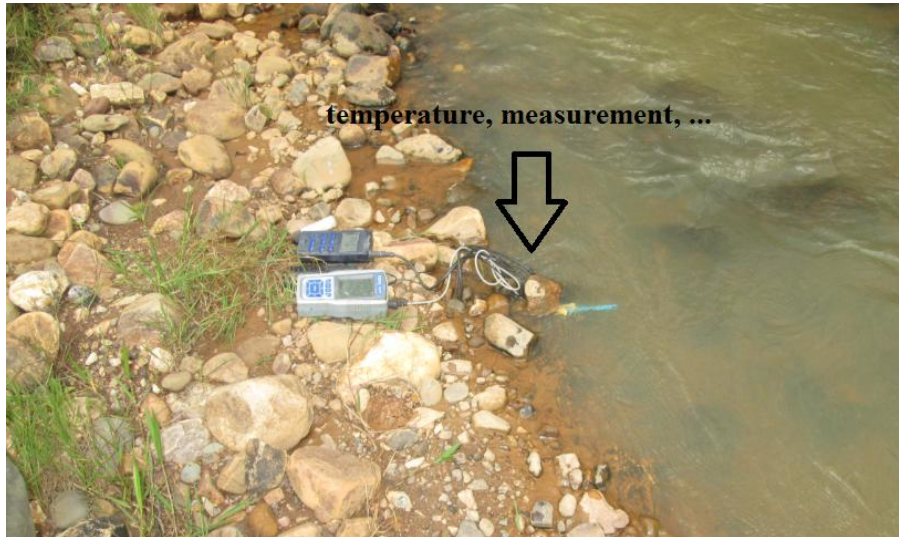


Figure24 : Electrical conductivity and, temperature measurement

➤ **Computational procedures of data from the site**²⁵

A floating method measurement is the summation of the products of the subsection areas of the stream cross-section and their respective average velocities. The continuity equation is used:

$$q = a * v \quad \text{Equation(2.5)}$$

$$Q = \sum_i^n a * v \quad \text{Equation(2.6)}$$

Where: q: is the discharge from an individual section;

a: is an individual section area;

v: is the mean velocity of the flow normal to the section; and

Q: is the total discharge from the cross-section.

Using the mid -section method of computing a floating measurement method, it is assumed that the velocity sample at vertical represents the mean velocity in a rectangular subsection.

The cross-section is defined by depths at verticals 1, 2, 3, 4 . . . n in Figure 25. At each vertical, the velocities are sampled using a floating object and a stop watch to obtain the averaged mean velocity of each section. The subsection discharge is then computed for any section at vertical X by use of the equation,

$$q_x = V_x \left(\frac{(b_x - b_{(x-1)})}{2} + \frac{((b_{(x+1)}) - b_x)}{2} \right) d_x \quad \text{Equation (2.7)}$$

$$q_x = \frac{V_x (b_{(x+1)} - b_{(x-1)})}{2} \times d_x \quad \text{Equation (2.8)}$$

Where:

q_x = discharge through section X;

V_x = mean velocity at vertical X;

b_x = distance from initial point to vertical X;

$b_{(x-1)}$ = distance from initial point to preceding vertical;

$b_{(x+1)}$ = distance from initial point to next vertical;

$d_{(x)}$ = depth of water at vertical X.

1.2.3.....n Observation verticals

$b_1, b_2, b_3, \dots, b_n$ Distance in centi-meters from the initial point to the observation vertical

$d_1, d_2, d_3, \dots, d_n$

Depth of water in centi-meters at the observation vertical

Boundaries of observations

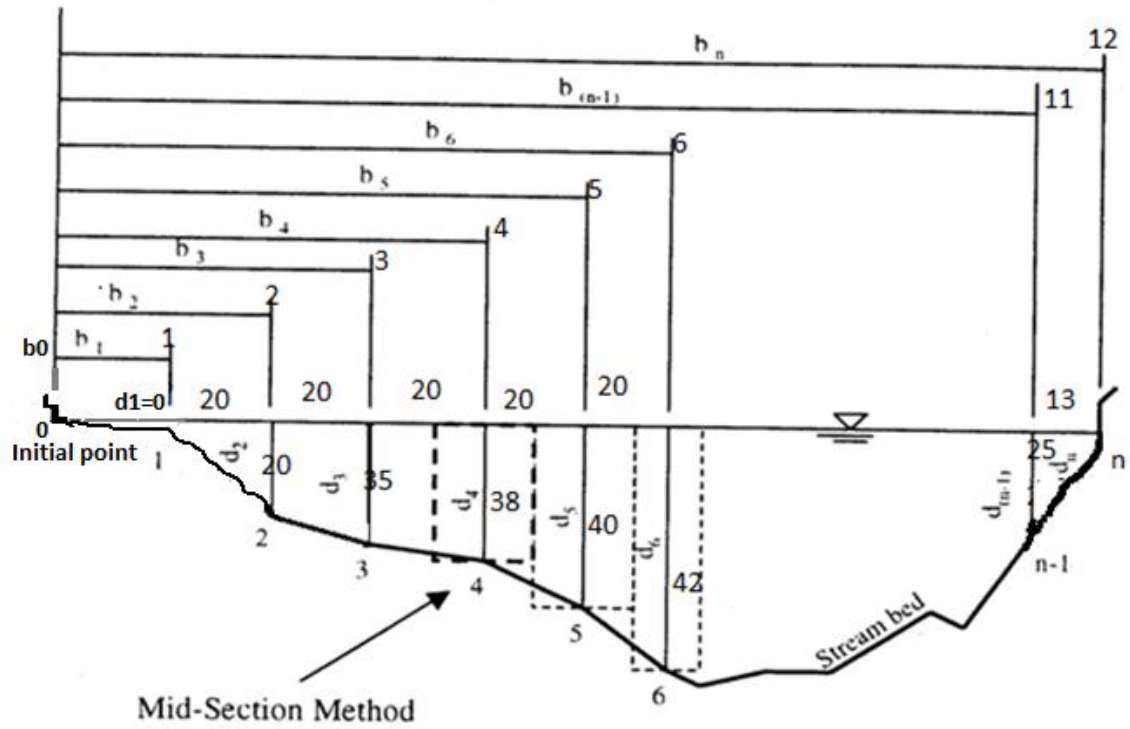


Figure 25 Midi-section method of computing cross-sectional area for floating measurements method.

$$q_1 = V_1 \left[\frac{b_2 - b_1}{2} \right] d_1$$

Equation (2.9)

$$q_n = V_n \left[\frac{b_n - b_{(n-1)}}{2} \right] d_n$$

Equation (2.10)

Looking at Figure 25, Equation 2.9 would be zero because $d_1=0$. Therefore, the small area defined by Equation 2.7 is assumed to be zero, which is commonly accepted because the discharge would be very small; but, in some cases this is not true when there is some velocity near the bank.

For the last section depicted in Figure 25 and represented by Equation 2.9, the velocity V_n cannot be measured, so it has to be estimated using the procedure of velocity at vertical walls.

Table 5. Site survey data calculation

GISUMA River Micro Hydro Power Plant					
GISUMA RIVER FLOW CALCULATION					
Point No	D:distance (m)	d:depth (m)	v = (m/s)	A (m ²)	Q(m ³ /s)
1	0.00	0.00	0.00	0.000	0.000
2	0.20	0.20	0.541	0.04	0.022
3	0.40	0.35	0.721	0.07	0.051
4	0.60	0.38	0.832	0.076	0.063
5	0.80	0.40	0.743	0.08	0.059
6	1.00	0.42	0.772	0.084	0.065
7	1.20	0.54	0.770	0.108	0.084
8	1.40	0.46	0.724	0.092	0.067
9	1.60	0.42	0.691	0.084	0.058
10	1.80	0.36	0.694	0.072	0.050
11	2.00	0.25	0.392	0.05	0.042
12	2.13	0.00	0.000	0.000	0.000
Average velocity (m/s)			0.688		
Total discharge (m³/s)					0.561

Temperature (°C)	27.60
Electrical Conductivity (µs/cm)	68.60
Segment Discharge= $A_N V_N$ from equation 2.5	

5.2. Penstock Hydraulic Calculations.

➤ Internal diameter penstock calculation³⁰

Practical and empirical equations used to determine the diameter of a penstock is given by equation (3.1) as shown in Figure 26, Using the head loss condition⁵,

$$h_L = \frac{V^2 L n^2}{R^{4/3}} \leq 0.005 H_{gross} \quad \text{Equation (3.1)}$$

Ludin – Bundschu has given empirical equations to compute the economical pipe inner diameter by depending on the head shown in the Figure 26,

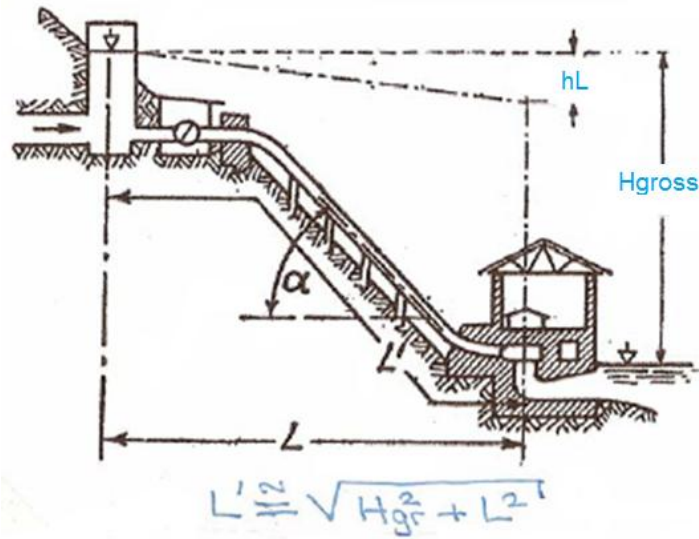


Figure 26 : Gross head

$$H_{gross} < 100m \rightarrow D = (0.05Q^3)^{1/7} \text{ m} \quad \text{Equation (3.2)}$$

$$H_{gross} > 100m \rightarrow D = [(5.2Q^3)/H_{gross}]^{1/7} \text{ m} \quad (\text{Bulu, 2000}) \quad \text{Equation (3.3)}$$

As $Q = 0.561 \text{ m}^3/\text{s}$ and $V = 0.688 \text{ m/s}$ from site survey calculation (Table 5).

Applying the Equation (3.3) as $H_{gross} = 131.5 \text{ m}$

$$D = [5.2 * 0.561^3 / 131.5]^{1/7} = 0.51 \text{ m} = 510 \text{ mm} \quad \text{Equation (3.4)}$$

The internal diameter of pipe is **510 mm**

Q = design flow in m^3/s

V = velocity of the water flow (m/s)

D = Internal diameter of pipe in mm

From the Figure 26, the slope angle of the penstock will be assumed as $\alpha = 45^0$ and $L = 131.4\text{m}$,

The length of the penstock will be, $L' = \sqrt{L^2 + H_{gross}^2} = \sqrt{131.4^2 + 131.5^2} = 186\text{m}$

Manning coefficient = $n = 0.014$ (Penche, 2004) from the Appendix A3, if we consider the penstock fabricated to concrete (steel forms smooth finish).

➤ The wall thickness in case of single penstock calculation:

The wall thickness in case of single penstock arrangement is determined by using the equation⁵:

$$e = \frac{pD}{2\sigma_{steel}} \quad \text{Equation (3.7)}$$

Where,

p = static + water hammer pressure

σ_{steel} = tensile stress of the steel

D = diameter of the penstock,

V = flow velocity,

e = wall thickness,

G = weight of the penstock.

5.3. Water Hammer

When a liquid flowing in a pipeline is abruptly stopped by the closing of a valve, dynamic energy is converted to elastic energy and a series of positive and negative pressure waves travel back and forth in the pipe until they are damped out by friction. This phenomenon is known as water hammer²⁷.

The pressure p_h caused by water hammer is expressed as:

$$p_h = \rho C_p V \quad \text{Equation (3.8)}$$

Where C_p is the velocity of a pressure wave

$$\rho_{\text{water}} = 1000 \text{ kg/m}^3$$

The total pressure at the valve immediately after closure is,

$$p_{\text{total}} = p_h + p \quad \text{Equation (3.9)}$$

Assume t is the time that a positive pressure will be maintained at the valve, then

$$t = 2L/C_p \quad \text{Equation (3.10)}$$

The velocity of a pressure wave in a water pipe usually ranges from 600 to 1200 m/sec for normal pipe dimensions and materials.

Calculation of Steel Pipe Thickness

The minimum thickness of steel pipe of penstock is determined by following formula.

$$t_0 = \frac{P \times d}{2 \times \theta_a \times \eta} + \delta t \quad (\text{cm}) \quad \text{and} \quad t_0 \geq 0.4 \text{cm} \quad \text{or} \quad t_0 \geq (d+80)/40 \text{ cm}$$

where, t_0 : minimum thickness of pipe

P: design water pressure i.e. hydrostatic pressure + water hammer (kgf/cm^2) , in micro-hydro scheme $P=1.1 \times \text{hydrostatic pressure}$.

for instance, if the head (H_p , refer to following figure) which from headtank to turbine is 25m, $P=2.5 \times 1.1=2.75 \text{ kgf/cm}^2$.

d: inside diameter (cm)

θ_a : admissible stress (kgf/cm^2) SS400: 1300 kgf/cm^2

η : welding efficiency (0.85~0.9)

δt : margin (0.15cm in general)

From the above formula, where: Head=13.5m $\rightarrow P=13.15 \times 1.1 = 14.465 \text{ kgf/cm}^2$

$$d=510 \text{ mm}$$

$$t_0 = \frac{P \times d}{2 \times \theta_a \times \eta} + \delta t$$

$$=[(14.465 \times 510)/(2 \times 1300 \times 0.85)] + 0.15 = 0.48 \text{ cm}$$

The penstock weight in case of single penstock installation is calculated by equation:

$$G = \gamma_{\text{steel}} \pi D e g \quad \text{Equation (3.15)}$$

The weight of the L length penstock with specific mass (density) of steel $\gamma_{\text{steel}} = 7800 \text{ (kg/m}^3\text{)}$ is,

$$G = 7800\text{kg}/\text{m}^3 \cdot \pi \cdot 0.510 \cdot 0.48 \cdot 9.81 \cdot 100 = 5870861\text{kg} = \mathbf{5880.861\text{tons}}$$

5.4. Net Head Calculation²⁷

Once the gross head has been measured, an applicable factor must be determined to allow for losses through the penstock to calculate the net head available at the site.

The net head is the gross head minus the applicable losses (the head loss, h_L).

The head loss includes the loss at the intake entrance, loss due to the gate pier, and other various losses in the settling basin, channel, head tank, penstock, draft tube tailrace and losses due to change in cross sectional areas and bend of channels. Among these head losses, the greater is the friction head loss in the penstock which is being calculated in our report.

Due to friction, useful energy and pressure are ‘lost’ or ‘dissipated’ when a fluid flows through pipes.

Bernoulli’s equation indicates that the energy balance for an incompressible flow should be maintained as follows:

$$P_1 + \rho gh_1 + \frac{\rho v_1^2}{2} = P_2 + \rho gh_2 + \frac{\rho v_2^2}{2} \quad \text{Equation (3.17)}$$

Where:

P = pressure (Pa)

ρ = density (kg/m^3)

g = acceleration due to gravity ($9.81 \text{ m}/\text{s}^2$)

h = height (m)

v = velocity (m/s)

The energy balance will be maintained in a pipe (penstock) but energy will be lost in proportion to the inner surface of the pipe as defined by the Reynolds number.

The value of the Reynolds number therefore depends on the temperature of the water, the site conditions, the penstock diameter and the water flow which can be calculated from values of V (measured mean speed of the water), D (measured diameter of the penstock) and kinematic viscosity of water ν .

The Reynolds number “ R ” is defined as:

$$R = V \cdot \frac{D}{\nu} \quad \text{Equation (3.18)}$$

Where V = Mean speed of the flow (m/s)

D = A nominated characteristic length of the system (in the case of a hydropower penstock, the diameter), ν = the kinematic viscosity of the fluid

The kinematic viscosity of water is temperature dependant and the viscosity of the water flowing through a penstock will also depend on the nature and the characteristics of the catchment area (soil content, pollution, etc.). A value of kinematic viscosity of water is determined considering the temperature of water measured on the site:

Calculation:

$$R = 0.688 \cdot 0.51 / \nu$$

$T = 27.6^{\circ}\text{C}$ (from the site survey Table 5)

The kinematic viscosity will be calculated by interpolation of the value which corresponds to 20°C and 30°C (Appendix A4) as follows³⁰:

$$v_2 = \frac{(T_2 - T_1)(v_3 - v_1)}{T_3 - T_1} + v_1 \quad \text{Equation (3.19)}$$

$$= \frac{(27.6 - 20)(0.801 - 1.004)}{30 - 20} + 1.004 = 0.849 \times 10^{-6} \text{ m}^2/\text{s}$$

$$R = 0.688 \times \frac{0.51}{v} = 0.688 \times \frac{0.51}{0.849} \times 10^{-6} = 2.6 \times 10^5$$

Both the friction coefficient and the Reynolds number must be determined to calculate the head loss. The friction coefficient depends on the height of the surface bumps in the pipe (e) relative to the diameter of the pipe (D). This is termed “Relative Roughness” and is calculated from e/D where “ e ” is the roughness height of the pipe and D is the diameter of the pipe. A value of “ e ” is determined by reading the table provided in Appendix A4

Once e/D (relative roughness) is determined, the friction coefficient can be obtained from the “Moody Diagram” as shown in Figure 27.

Reynolds number of 2.6×10^5 and calculated relative roughness is $e/D = 0.048/0.51 = 0.0943$ and a friction coefficient (f) is determined by following the curved line from the relative roughness value (on RHS) to the intersect of the Reynolds Number line and from this point, a line is projected to the left to read the friction (resistance) coefficient.

Moody Diagram

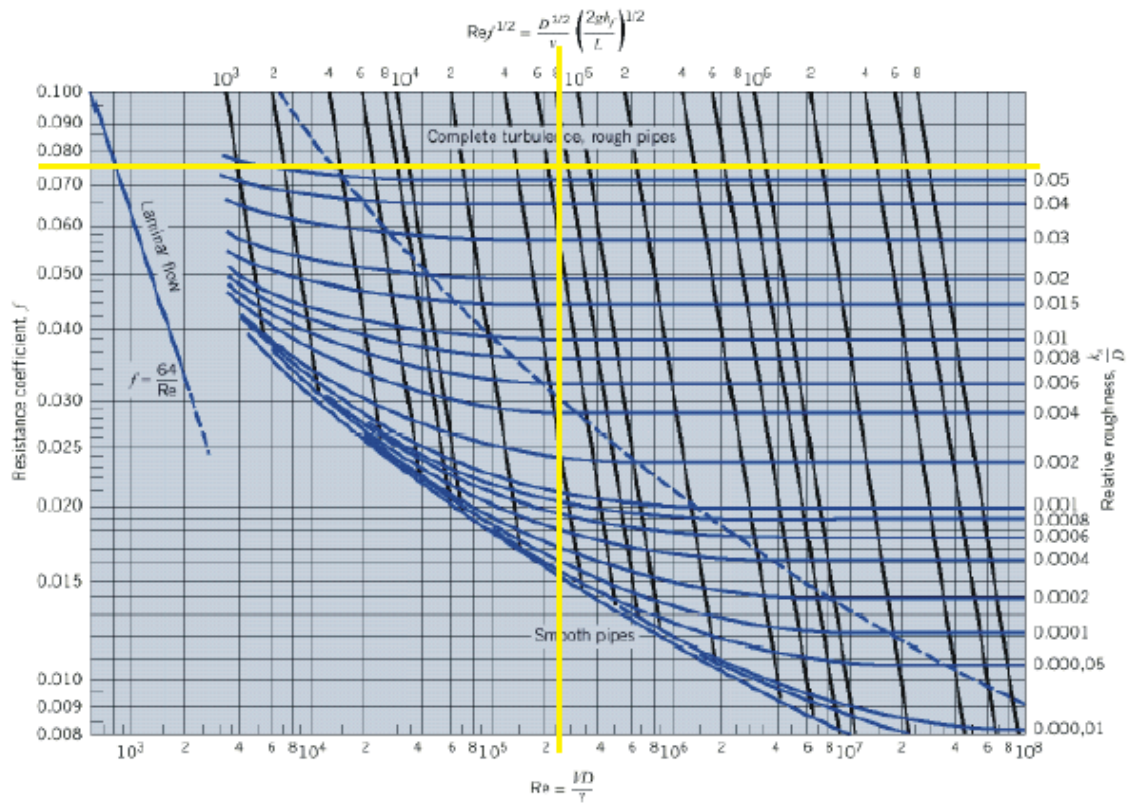


Figure 27 source :Moody Diagram to determine friction factor (Penche, 2004).

➤ Friction factor calculation:

It is often effective to solve for this friction factor using the above Moody Chart as follows²³:

1. The line referring to our relative roughness is on the right side of the diagram. In that case we have a printed line (correspond to 0.06).

2. This line to the left curves up until to reach the yellow vertical line, where corresponding to my flow's Reynolds Number (2.6×10^5)
3. This point is marked on the Chart.
4. Using a straight edge, the point straight left, is followed parallel to the x axis, until one reaches the far left side of the chart.
5. We read off the corresponding friction factor which is 0.077

Or as shown in yellow line, a calculated Reynolds number of 2.6×10^5 and calculated Relative Roughness of 0.06 would mean a friction factor (f) of 0.077. The friction coefficient is determined by following the curved line from the Relative Roughness value (on RHS) to intersect of the Reynolds Number line and from this point, a line is projected to the left to read the friction factor³⁰.

The head loss due to friction is then determined from the following³⁰ “Darcy-Weisbach” Formula:

$$H_f = f \left(\frac{L}{R} \right) \times \left(\frac{V^2}{2g} \right) \quad \text{Equation (3.20)}$$

Where: H_f = head loss (m)

f = friction factor

L = length of pipe work (m)

R = hydraulic radius of pipe work (m)

V = velocity of fluid (m/s)

g = acceleration due to gravity (m/s)

It should be noted that “for a round pipe with full flow the hydraulic radius is equal to ¼ of the pipe diameter” and the head loss due to friction may also be calculated from the following formula ³⁰.

$$H_f = \frac{2fLV^2}{Dg} \quad \text{Equation (2.21)}$$

Where:

H_f = head loss (m)

f = friction factor

L = length of pipe work (m)

D = inner diameter of pipe work (m)

V = velocity of fluid (m/s)

g = acceleration due to gravity (m/s)

Knowing the friction factor from the figure 27 we can calculate the energy losses in the penstock and bends as follow:

$$H_f = \frac{2fLV^2}{Dg} = 2 \times 0.077 \times 131.5 \times \frac{\frac{1.2384^2}{0.510}}{9.81} \cong 6.2m$$

³⁰The losses for the bend is 5% of $H_f = 6.2 \times 5/100 = 0.31m$

The net head is $131.5m - 6.2m - 0.31m = 125m$

$$\text{Penstock efficiency } n_{pen} = \frac{H_{net}}{H_{gross}} \quad \text{Equation (3.16)}$$

$$\eta_{pen} = \frac{125.3}{131.5} \times 100 = 95.28\%$$

5.5. Calculation of Power Output

Once the net head (H) and water flow rate (Q) have been determined, the calculation of available power (P) is straightforward and can be established from the formula ³²

$$P = P_0 \eta_{tg} = 9.81QH\eta_{tg} \text{ (kW)} \quad \text{Equation (3.22)}$$

Where, P = Power output, η_{tg} = Combined efficiency of generating equipment = Turbine efficiency (η_t) $\times$ Generator efficiency (η_g) .

$$P = 9.81 \times 0.561 \times 125.3 \times 0.72 = 451 \text{ kW}$$

³²The electricity will be continuously generated the whole year at the output of P(kW),

$$\text{Annual Energy generation} = P \text{ (kW)} \times 24\text{h/day} \times 365\text{days} \times 0.85 = 8760 \times P \text{ (kWh)} \times 0.85$$

$$= 451 \times 8760 \times 0.85 = 3,358,146 \text{ kWh} \quad \text{Equation (3.22)}$$

5.6. Barrage Options

The first option suggests a barrage at the very beginning of the rapids and the barrage height is kept to about 3m to provide enough height to send water to a channel resting on the riverbank. Necessary water depth above the entrance to the channel is maintained on top of storage reserved for accumulation of the sediments ⁸

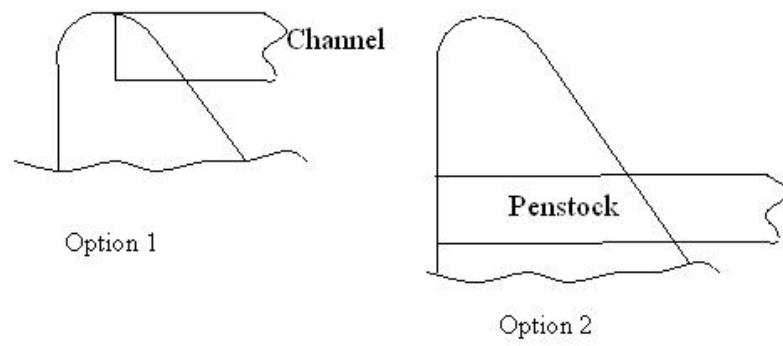


Figure 28: source: Options on intake at the barrage (EWSA, 2011).

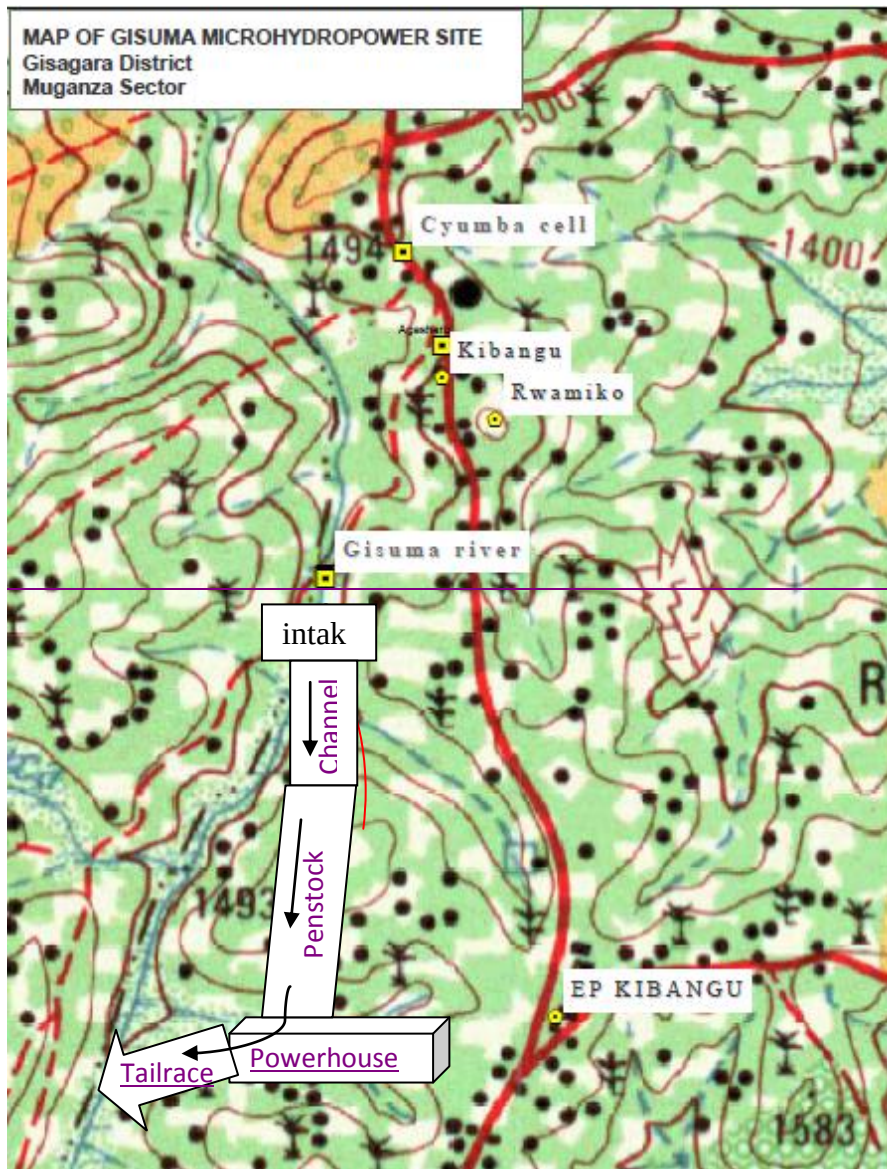


Figure29. Map taken from GPS

Figure 20 shows the options that can be considered at the barrage for intake options. The option 1 is to start a channel at the barrage and connect it to a penstock through a forebay tank. The option 2 is to start the penstock at the barrage and this will eliminate the forebay tank but may increase the cost as the length of the penstock will increase⁸.

If the option 1 type of intake to a channel is considered the water depth required at the barrage is only to head up the water to facilitate water flow into the channel at the required rate. Therefore the depth can be lesser depending only on the required channel bed level. Riverbanks at the site are about 1 m high. In order to support the channel on the riverbanks the barrage minimum height needed is about 1m. ^[8]

If the option 2 is considered the water depth required at the barrage is much higher compared to that of option 1. The requirement here is to provide enough depth to prevent vortex formation and air suction into the penstock. The depth will therefore be about 3 m in this case leaving about 1 m depth storage for sediments, as the sediment flow is moderate²⁵.

These two options were considered for the site. The reason for the two locations is to select the one with the best geological conditions with economically viable project. The doubt about the upstream location is raised, as a longer conveyance system is needed. The upstream location will result in additional excavations ⁸.



Figure30: Picture showing the location for the barrage

5.7. Options on Powerhouse Location

For the maximum utilization of the available head the powerhouse should be located towards the end of the steep riverbed slope. Therefore, best location of the powerhouse is seen on the right bank closer to the end of the river rapids say at position A(fig22). The powerhouse is located little away from the river and in turn at a higher elevation compared to the riverbed to avoid inundation due to floods ¹⁷.



Figure 31. Picture showing the proposed location of power house

5.8. Selection of GISUMA MHP Equipment.

5.8.1. Powerhouse

In a small or micro hydropower scheme the role of the powerhouse is to protect the electromechanical equipment that convert the potential energy of water into electricity. The

number, type and power of the generators, their configuration, the scheme head and the geomorphology of the site determine the shape and size of the building.

As shown in Figures 32, the following equipment will be displayed in the powerhouse:⁸

- ✓ Inlet gate or valve
- ✓ Turbine
- ✓ Generator
- ✓ Control system
- ✓ Condenser, switchgear
- ✓ Protection systems
- ✓ Standby Diesel generator
- ✓ Etc.

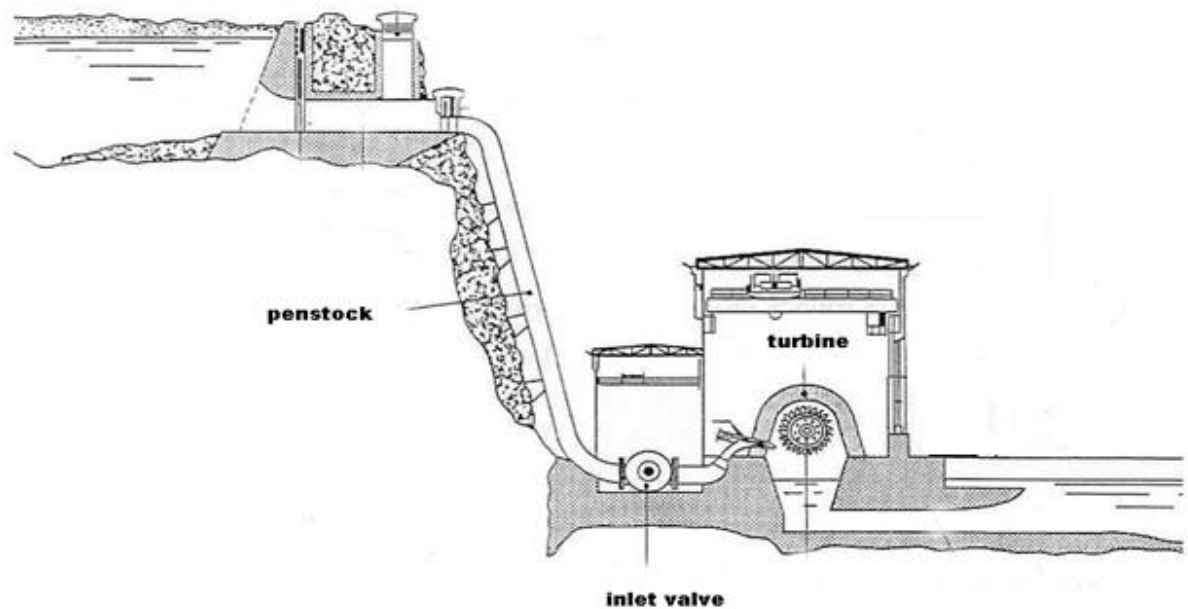


Figure 32 : Schematic view of a powerhouse –high and medium heads

In medium and high head schemes, powerhouses are more conventional (Figure 32) with an entrance for the penstock and a tailrace. Although not usual, this kind of powerhouse can be underground.

5.8.2. Hydraulic turbines

The purpose of a hydraulic turbine is to transform the water potential energy to mechanical rotational energy. Although it is appropriate to provide a few criteria to guide the choice of the right turbine for a particular application and even to provide appropriate formulae to determine its main dimensions. It is necessary to emphasize, however, that no advice is comparable to that provided by the manufacturer, and every developer should refer to manufacturer from the beginning of the development project.

The turbine is chosen by considering the turbines' type field of application chart shown below in yellow colors¹⁵

$$\text{Turbin runner diameter} = 0.482Q^{0.45} = 0.482 \cdot 0.562^{0.45} = 0.372\text{m}$$

Where Q is the discharge in m³/min

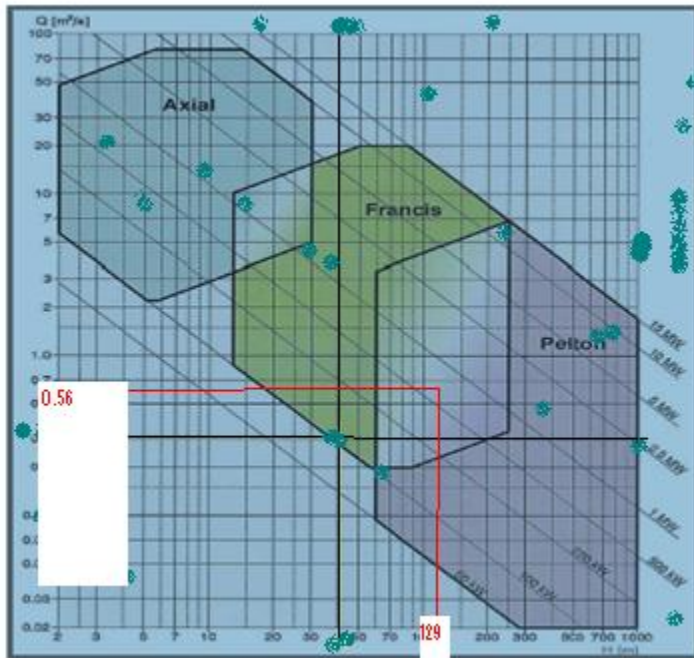


Figure 33 Turbines' type field of application chart

The rated flow and net head determine the set of turbine types applicable to the site and the flow environment. Suitable turbine is that for which the given rated flow and net head plot within the operational envelopes (Figure 33). A point defined as above by the flow and the head in yellow color will usually plot and provide the suggested turbine to those specific parameters values. All of those turbines are appropriate for the job, and it will be necessary to compute installed power and electricity output against costs before making a decision. It should be remembered that the envelopes vary from manufacturer to manufacturer and they should be considered only as a guide.

➤ Turbine Supply Suggested.

In our project, Horizontal shaft Francis turbine with all accessories: runner diameter: **37 cm**, rated **flow: 0.58m³/s**. is selected (fig.34).

Francis turbines are reaction turbines, with fixed runner blades and adjustable guide vanes, used for medium head. In this turbine the admission is always radial but the outlet is axial. Figure 34 shows a horizontal axis Francis turbine. Their usual field of application is from 25 to 350 m head. Francis turbines can have vertical or horizontal axis, this configuration being really common in small hydropower plant.



Figure 34 : Horizontal axis Francis Turbine

Turbine's type supply company

There are several manufacture's company of turbines and the popular is OSSBERGER, Chinese turbine, etc...

OSSBERGER TURBINE

In almost 120 countries more than 10000 Ossberger turbines are presently producing energy from hydro power in an environmentally friendly way.

Brief description of ossberger turbine

Some main parts of the Ossberger turbine;

Guide Vanes:

The subdivided guide vanes provide a flat efficiency curve, they need to direct the water for a jerk-free entry into the runner. As the manufacture of cast guide vanes would demand too large tolerances calibrated profiled guide vanes are referred to. Both guide vanes can be set up independently, they can easily be dismantled towards the radial direction without special tools, the corresponding forces are absorbed by maintenance- free friction bearings brand Perm glide with shaft protecting sleeves.

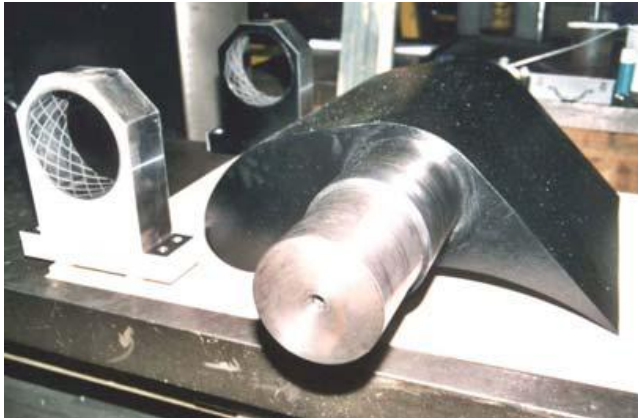


Fig.35.Turbine guide vanes.

Bearings:

The OSSBERGER turbine is equipped with standardized self-aligning roller bearings, designed for an infinite service life. The bearing casings and the bearing inserts can be considered as one unit, they are fixed in the turbine casing by a locating bunch. This will permit to dismount the runner radially without removing the bearing casings from the runner shaft. A simple readjustable gland packing arrangement with hemp tallow cord seals the shaft.

Draft Tube:

The draft tube is essential for utilizing the level difference between runner and downstream water level. During turbine operation the air in the casing is taken along with the discharged water. Thus a vacuum is formed; for the exterior atmospheric pressure the suction column rises. A simple venting valve which is free of own friction controls the vacuum in the turbine casing to utilise the energy potential optimally.

Runner:

The cylindrical runner core consists of conservatively designed cam disks and, following its width, several intermediate disks to which the profiled blades are adapted and welded. This will make the runner extremely solid by stiffening it at the same time in such a way that no vibrations are faced. The blades consisting of bright- drawn profiled steel mean an ideal solution regarding solidity and water guidance. By utilizing the bright-drawn and exact blade profiles an ideal balance condition is achieved automatically, only slight corrections are required on the balancing machine. The blades bent in a linear way only do not produce any axial thrust, thus there is no need for pressure bearings. Another advantage of the flow guidance is that leaves, grass or snow, pressed between the blades when the water enters the runner, are spilled out again after a half turn by the leaving water, backed-up by the centrifugal force. Thus the self-cleaning runner is never obstructed.



Fig.36. Turbine runner.

Base Frame:

A stable base frame is provided between turbine and foundation, permitting a rapid and sure installation.

Regulator:

For an automatic operation of the machine unit and the turbine regulation in accordance with the water level an electro-hydraulic turbine controller has been foreseen. It is composed of a hydraulic unit and a control switchboard. Emergency stop in case of mains failure or generator switch release are made without any foreign energy through storage weights. The continuous

water level registration by a depth gauge and the regulation which is continuous either mean a precondition for an almost constant upstream water level and thus an optimal utilization of the existing flow. The computer capacity of the specially adapted regulator element has been overdimensioned and allows for an enlarged servicing and observance as well as for the necessary plain text display, and all this operator-controlled, at the device directly. No programming knowledge nor devices are necessary.

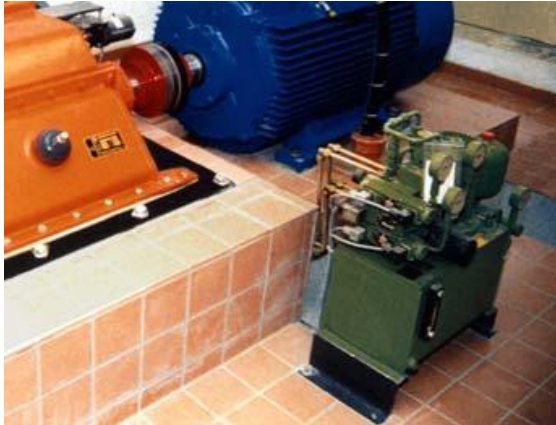


Fig.37. *Regulator unit.*

Installation and commissioning:

The proposed concept does not comprise any elements that need to be concreted in advance. In case of order provided, apart from the plan of installation, with a plan of foundations which will enable to prepare the site for a quick and trouble-free erection. The turbine and the base frame are lodged on the carriers laid in the primary concrete which is situated above the draft tube pit. Own bases have been foreseen for gearbox and generator. The fixation is made by means of foundation blocks.

Each plant is completely assembled prior to its dispatch and tested at the manufacturer's works. Thus it is ready for operation at site as soon as the erection work has been finished. The equipment is delivered as premounted units. Thus erection is limited to the exact alignment of the elastic couplings.

Technically, Ossberger turbine offer the following advantages:

1. The flanged corner casing can be easily removed;
2. The runner can be disassembled flow direction even in very confined spaces;
3. The guide vane bearings are easily replaced and virtually maintenance free;

4. The runner blades are hydraulically designed and made of cold drawn pre-formed steel;
5. The preassembled machinery components are easily installed;
6. The guide vane are hydraulically efficient and made of the highest quality steel;
7. The tightly closing turbine guide vanes eliminate the need for additional water intake gates and manually activated service valve in sufficient even at high head;
8. The stable efficiency over a wide range of inflows; from 17% to 100% of design flow;

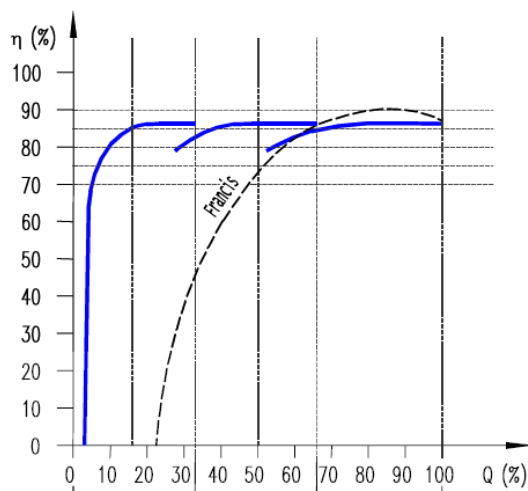


Fig.38. Ossberger's Discharge/Efficiency curve.

9. The cavitation free-operation from zero to full load;
10. The heavy continuous guide vane shaft, which is solidly welded to the guide vane.

For these reasons Ossberger has been for decades the technological leader in the field of small scale hydro power for its customers' benefit and this has been perfected by using standardized components.

Ossberger turbine's initial cost is high compared with other type of company's turbine but its running cost is low as it last longer with less maintenance and less spare parts. As an example a one set of OSSBERGER high quality Francis turbine for hydropower plant of 300kW costs US\$200,000 while the Chinese 400KW high quality francis turbine costs US \$100,000/set.^[34] That is why in our thesis project we decided to use Chinese turbines to reduce the initial cost of investment.

5.8.3. Generators

Generators transform mechanical energy into electrical energy. Although most early hydroelectric systems were of the direct current variety to match early commercial electrical systems, nowadays only three-phase alternating current generators are used in normal practice. Depending on the characteristics of the network supplied, we can choose:

➤ Synchronous generators

They are equipped with a DC electric or permanent magnet excitation system (rotating or static) associated with a voltage regulator to control the output voltage before the generator is connected to the grid. They supply the reactive energy required by the power system when the generator is connected to the grid. Synchronous generators can run isolated from the grid and produce power since excitation is not grid-dependent.

The synchronous generator is started before connecting it to the mains by the turbine rotation. By gradually accelerating the turbine, the generator must be synchronized with the mains, regulating the voltage, frequency, phase angle and rotating sense²⁶. When all these values are controlled correctly, the generator can be switched to the grid. In the case of an isolated or off-grid operation, the voltage controller maintains a predefined constant voltage, independent of the load. In case of the mains supply, the controller maintains the predefined power factor or reactive power.

➤ Generator Supply Suggested:

Synchronous generator: rated capacity 451 kW, 380 V at 0.8 power factor, 1000 rpm, continuous load, with suitable bearing, excitation system and all associated auxiliaries required for coupling to horizontal shaft Francis turbine.

5.8.4. Switchgear Equipment

In many countries, the electricity supply regulations place a statutory obligation on the electric utilities to maintain the safety and quality of electricity supply within defined limits. The independent producer must operate his plant in such a way that the utility is able to fulfill its obligations. Therefore various associated electrical devices are required inside the powerhouse for the safety and protection of the equipment.

Switchgear must be installed to control the generators and to interface them with the grid or with an isolated load and also to provide protection for the generators. The generator breaker, either air, magnetic or vacuum operated, is used to connect or disconnect the generator from the power grid. The generator control equipment is used to control the generator voltage, power factor and circuit breakers.

5.9. Desilter

The lateral intake provides a good protection against bed-load entrainment, as well as, against floating debris, by means of the flushing facility, the forefront wide trashrack and the counter-gradient (4%) at the intake. Nevertheless, suspended sediments transported in the river flow can easily enter the approach channel and silt along the headrace channel or eventually reach the penstock intake.

Suspended sediments, especially particles of hard material such as quartz, which abound in Rwanda rivers, can affect turbine components by abrasion. In order to avoid turbine wearing or uncontrolled sediment deposition along the headrace channel, the intake structure will be equipped with a desilter.

The severity of abrasion and damages to hydro-equipments is a function of several variables, notably sediment hardness, shape, concentration and size, among which the latter plays the main role in the desilter design. The desilting efficiency is indeed defined as the design particle diameter which is trapped and flushed by the facility.

The control of turbine wear issues due to silt abrasion requires comprehensive design approach in which sediment properties, turbine mechanical and hydraulic design, material selection and equipment maintenance are all considered. The maintenance frequency for a standard Francis/Pelton turbine is generally of 6-7 years for efficiency of 0.2 mm, of 3-4 years for efficiency of 0.3 mm and 1-2 years for efficiency of a few 0.5 mm. Nevertheless, cost of the

desilter increases with its efficiency and in most of the conditions, selection of a design particle diameter lesser than 0.2 mm would results unpractical and excessively costly.

Based on the site condition, the turbine characteristics (Francis type), as well as the balance between cost of maintenance and cost of construction, the optimal efficiency for the Gisuma desilter has been selected as a low 0.2 mm, which is also commonly recommended by the engineering best practice for standard conditions in low and medium head plants, for good water quality, for medium/high quartz particles concentration.

5.9.1.Desilting Basin

Once selected the desilter efficiency, the main features of the desilting basin are defined based on the design discharge. The design methodology is as follows:

- The flow velocity in the basin should not exceed the critical velocity which would cause entrainment of the material already settled out to the basin bottom, thus:

$$U \leq a\sqrt{d}$$

where: U = recommended flow velocity (m/s),

d = target sediment diameter (mm),

$a = 0.44$, for $1 \text{ mm} > d > 0.1 \text{ mm}$.

- The basin cross-section should be designed to provide the recommended flow velocity U , thus:

$$B = \frac{U}{h}$$

where B is the basin width (m).

- The basin length should be appropriate to allow the suspended sediments to fall within the basin, thus:

$$L \geq \frac{hU}{u - w}$$

where: L = basin length (m),

h = basin height (m),

u = settling velocity of supposed spherical particles, i.e. by Stokes formula:

$$u = \frac{1}{18} \frac{g}{\mu} (\rho_s - \rho) d^2$$

where: μ = water absolute viscosity, 0.0012 kg/m s (27.6°C),

ρ_s = sediment density, 2600 kg/m³,

ρ = water density, 1050 kg/m³, relatively high turbidity,

g = acceleration of gravity, 9.78 m/s² (corrected for equatorial-latitude and 1700 a.s.l.),

w = adjust for effect of turbulence:

$$w = U \frac{0.132}{\sqrt{h}}$$

- In order to allow for the entire basin to contribute in the settling process, the length to width ratio should not be less than 4.0, therefore, the basin length is to be adjusted to fit with:

$$\frac{L}{B} \geq 4.0$$

In case the above relation can lead to an excessive basin length, the standard approach would suggest to sub-divide the desilter into two or several basins, each of them correctly proportioned.

- The basin bottom should consist of a mild grade toward the end section, in order to allow the settled sediments to move by gravity to the flushing facility. The recommended longitudinal and transversal grade are respectively in the order of 1:20 (H:V) and 1:7 (H:V). The bottom should be carefully treated with a surface finishing to reduce roughness as well as to prevent deterioration.

Table 6 -Gisuma – Desilter Basin.

Features	Design Parameter
Design discharge	0.561 m ³ /s
Desilter water level	1758.25 m a.s.l.
Desilter efficiency	0.2 mm
Flow velocity	0.688 m/s
Number of basins	1
Basin length	12.0 m
Basin height *	1.2 m
Basin width	2.8 m

* the additional height due to the sloping bottom has not been accounted in the effective basin height.

5.9.1.1.Desilting Basin Capacity

The desilting basin should be designed to provide a retention capacity capable to accommodate the expected volume of incoming sediments, which is a function of the sediment concentration in the river, without affecting the settling efficiency as well as with a very low maintenance requirement.

It is recommended that a long-term sediment sampling be carried out near the intake site to ensure that sufficient data is available for an optimal desilter design. Purpose of the sampling is to evaluate the mean sediment concentration together with other relevant sediment characteristics. The sampling program should extend through the entire rainy season and should comprise several daily readings. A minimum five year long collecting activity is always desirable. However, for mini-hydro project, where resources and time are not available to

undertake a comprehensive sampling program, selection of design parameters mainly depends on engineering judgment, supplemented by site observations.

Empirical formulae can be used to support decision making, the following proposed by Garde & Kothyari (1985) is commonly adopted and returns reliable estimate of the mean sediment load:

$$V_s = 530.0 P^{0.6} F_e^{1.7} S^{0.25} D^{0.1} \left(\frac{P_{max}}{P} \right)^{0.19}$$

where: V_s = mean sediment load (t/km²/year),

s = average river slope, 0.039 m/m,

D = drainage density, as total stream length divided by estimated catchment area, 0.16 km/km²,

P = mean annual precipitation, 125mm,

P_{max} = average precipitation for wettest month, 18.5 mm,

F_e = ground cover factor, 0.4 for grass and cultivated land area.

The mean sediment concentration is thus obtained by the following:

$$C_c = \frac{V_s A}{V_Q \rho_s}$$

where: C_c = mean sediment concentration (%),

A = estimated catchment area, 13.2 km²,

ρ_s = sediment density, 2.6 t/m³;

V_Q = annual discharge volume (m³/year), based on the average river discharge, 0.6 m³/s.

The above equation allows to compute the total sediment load (V_s), bed + suspended, where the bed-load represents normally the most of the sediment transport and the suspended load can be correctly estimated as a small 10-20 %. Suspended sediments represent the key input in the design of lateral intakes, as the bed-load is diverted away from the intake and flushed downstream from the diversion structure.

Table 7 - Gisuma – Desilter Retention Capacity.

Features	Design Parameter
Mean sediment load	530 t/km ² /year
Mean sediment concentration	0.046 %
Mean sediment volume	5 m ³ /day
Suspended sediment volume	0.8 m ³ /day (15%)
Desilting basin capacity	40.0 m ³
Basin capacity loss factor	-2% /day

The basin capacity loss factor is defined as the portion of the active settling volume which may be silted by the incoming sediments on a daily basis. Normally, when the basin capacity reduces below the 80% of the original design, the settling efficiency may be significantly compromised. According to the figures, by a rate of -2%/day loss, the Gisuma desilter efficiency would be affected after approximately 10 days of operation. In such condition, implementation of a manual sediment removal solution would result in being impractical, it is therefore recommended to install a completed flushing system.^[15]

5.9.2. Flushing System

Flushing systems can be essentially classified into two main categories: continuous flushing and intermittent flushing. Both the solutions comprise a discharge pipe, located on the bottom of the desilter end-section (one for each basin), designed to convey the mixture of water and sediments by pressurized flow, and a valve (sliding gate), which can be either manually or mechanically operated (hoist).

The intermittent flushing requires an extra storage capacity in the desilting basin in order to retain the volume of the incoming sediments between two flushing operations. Conversely, the continuous flushing allows to constantly and automatically remove the silted sediments, thus minimizing maintenance, inspection and avoiding costs for extra capacity. The disadvantage of the continuous flushing is related to the discharge of useful water, which would be no more available for power generation.

According to the volume of incoming sediments expected at the Gisuma desilter (see “basin capacity loss factor”), the intermittent flushing would require an excessive extra capacity which will likely result not cost-effective, or conversely, a tight flushing program (10 days schedule)

with impacts on the powerhouse operation. Therefore, the optimal solution is the implementation of a continuous flushing system. To minimize (or avoid) diversion of useful water through the discharge pipe, the flushing gate will be regulated to constantly release an amount water equal to the ecological flow, thus preserving the design discharge for power generation.

5.9.3. Sediment Removal Efficiency

The amount of sediments trapped by the desilting basin is generally expressed in terms of removal efficiency, which is defined as:

$$\alpha = \frac{(q_{si} - q_{se})}{q_{si}}$$

where: α = basin removal efficiency,

q_{si} and q_{se} = amounts of sediment entering and leaving the basin (m^3/s).

Several formulae are available in literature (Camp (1946), Dobbins (1944), USBR (1975)), which have been derived by laboratory data, however, they only cover the settling phase in basin operation. These methods cannot be directly applied when the desilting basin is hydraulically flushed continuously.

A recent study carried out by Raju & Kothiyari (1999) presents a new equation to check sediment removal efficiency, the approach is summarized as follows:

$$\alpha = 11.7 \left(\frac{u}{U} \right)^{0.81} \left(\frac{L}{h} \right)^{0.23} \left(\frac{h^{1/6}}{n\sqrt{g}} \right)^{0.98}$$

where: α = sediment removal efficiency (%),

u = settling velocity (m/s),

U = flow velocity in the basin (m/s),

L = basin length (m),

h = basin height (m),

n = Manning's coefficient for basin bottom, 0.013 (concrete).

Table 8 - Gisuma – Basin removal efficiency.

Features	Design Parameter
Basin removal efficiency	> 100%

The above result confirms accuracy of the desilting basin design, the entire amount of incoming sediments will accordingly be trapped and flushed out by the desilter.

Chapter6. Economic Analysis

6.1. Introduction

An investment in a small hydropower scheme will incur costs as well as earn income over the life of the project .The costs include a fixed component - the capital cost, insurance, taxes other than the income taxes, rates etc. and a variable component operation and maintenance expenses, salaries, income tax etc

The economic analysis is a comparison of costs and benefits that enables the investor/investors to make an informed choice whether to develop the project or abandon it. It is also possible that a choice may be made between different hydro projects so that the investment can be made in the one that gives the best return

From an economic viewpoint, a hydropower plant differs from a conventional thermal plant; because its initial investment cost per kW is much higher but the operating costs are extremely low, since there is no need to pay for fuel.

The economic analysis can be made by either, including the effect of the inflation, or omitting it. Working in constant monetary value has the advantage of making the analysis independent of the inflation rate. Valued judgments are easier to make in this way. Because the analysis refers cash flows to the present time it is easy to judge the overall value of the investment.

The estimation of the investment cost constitutes the first step of an economic evaluation. For a preliminary approach, the estimation can be based on the analyses the cost of the different components of a scheme like weir, water intake, canal, penstock, powerhouse, turbines and generators, transformers and transmission lines.

For the case of Gisuma MHP cost analysis we used the RETScreen software-Small Hydro projects, designed as a **unique decision support tool developed with the contribution of numerous experts from government, industry, and academia**. The software, provided free-of-charge, can be used worldwide to evaluate the energy production and savings, costs, emission reductions, financial viability and risk for various types of energy-efficient and Renewable-Energy Technologies (RETs) (RETScreen, 2004) and the obtained results are summerised in table 33. However as the used RETScreen software is given in French not in English depend on availability,the complete calculation process is found in appendix A5 to A9.

6.2. Cost analysis - Power project

Item	Cost(\$CAD)	Percentage
Feasibility study: .Detail cost estimation .Site inspection .Etc,...	162,000	3.12%
Development: . Licenses and permits .Etc,...	192,000	3.69%
Engineering: .Supervision of works	1,707,500	32.86%

Etc,....		
Electromechanical system:		
. Hydro-turbine , Generator and accessories		
.Equipment installation		11.06%
. Step-up transformer		
Sub total	574,795	
Civil engineering:		
.Road construction		
.Barrage		
.Canal		
.Penstock		27.72%
. Forebay		
.Transportation		
. Powerhouse		
Etc.....	1,440,374	

Other costs:		
. Training for technicians		
. Accommodation		21.55%
.Travels		
Etc,....	1,119,579	
Total investment cost	5,196,248	100%

Table 9.a

➤ **Annual costs (credits)**

Item	Cost (\$CAD)	Percentage
.Spare parts & Labour		
. Contingency costs		
.Etc,...	158,445	100%
TOTAL	220,077	22.16%

Table 9.b

➤ **Periodic cost (credits)**

Item	Cost(\$CAD)	Percentage
.Turbine maintenance		
Etc,...	99,307	10.00%
TOTAL	99,307	10.00%

Table 9.c

Table 9(a,b,c). Summary,s RETScreen cost analysis of Gisuma MHP.

6.3. Financial Analysis

An investment in a small hydropower scheme entails a certain number of expenses, extended over the project life, and procures some revenues also distributed over the same period. The expenses are referred to as fixed component, the capital cost, insurance, taxes, and a variable component like the operation and maintenance expenses.

In order to define “most economic turbine flow” and to limit the investment costs of plant, following issue were compared:

- available discharges and their duration, in terms of days/year, taking into account flow duration curve resulting from hydrologic analysis;
- annual power generation considering head losses due to hydraulics in the piping ;
- the efficiency of turbine;
- Total costs and revenues arising from energy annual production considering load and capacity factors analysis of the plant(s).

Optimal discharge was selected taking into account the values of reconnaissance survey measures indicated overleaf by comparison of different flow measurement methods of float. The net present value (NPV) and the pay-back period has been discussed also in this paper theoretically and quantitatively demonstrating how one interested could predict the time to recover the loan and earn of revenues.¹⁹

The NPV determines whether or not the project is generally a financially acceptable investment. Positive NPV values are an indicator of a potentially feasible project.

The net present value is equal to the present value of future value, discounted at the marginal capital cost minus the present value of the cost of investment. The difference between revenues and expenses, both discounted at a fixed, periodic interest rate, is the net present value (NPV) of the investment. The formula for calculating NPV, assuming that the cash flows occur at equal

time intervals and that the first cash flows occur at the end of the first period, and subsequent cash flow occurs at the end of subsequent period, is as follow:

$$(NPV)_n = \frac{C_n}{(1+d)^{n-1}}$$

where:

$(NPV)_n$ = *Net present value in a year*

C_n = *Net cost value (after subtracting credits from cost in year n) and,*

d = *discount rate*

levelized Net present value cost considering all the expenses,

$$NPV = \sum_{i=1}^{i=n} \frac{Ri - (Ii - Oi - Mi)}{(1+r)^i} + Vr$$

where:

Ii = investment in period i ;

Ri = revenues in period I ;

Oi = operating costs in period I ;

Mi = maintenance costs in period I ;

Vr = residual value of the investment over its lifetime, where equipment lifetime exceeds the plant working life;

r = periodic discount rate;

n = number of lifetime periods e.g. years, quarters, months etc.

By using RETScreen, the Net Present Value (NPV) of Gisuma MHP is 545,940 (\$CAD). (Appendix 9). In this case study of GISUMA MHP the NPV is positive means that the project is feasible.

The payback period method determines the number of years required for the invested capital to be recovered by resulting benefits. The required number of years elapsed to complete loan payment is termed as the payback period. The calculation is as follow:

Payback period = Investment cost / Net annual revenue

Investment costs are usually defined as capital costs spent on the following initially before the plant operation through civil works, electrical and hydro mechanical equipments while benefits are the resulting net yearly revenues earned from power selling less the operation and maintenance costs spent in a certain period of time (Appendix A7 & A8).

From RET Screen software-Small Hydro Project , the Payback period of Gisuma Micro Hydropower plant is 10.1 years and the amount of money to be earned per year is 82,511(\$CAD) as illustrated in figure 35 and the project life is 35 years (Appendix A9).

feasibility study of gisuma hydro power plant(Rwanda), Gisuma-Muganza-Gisagara-Rwanda

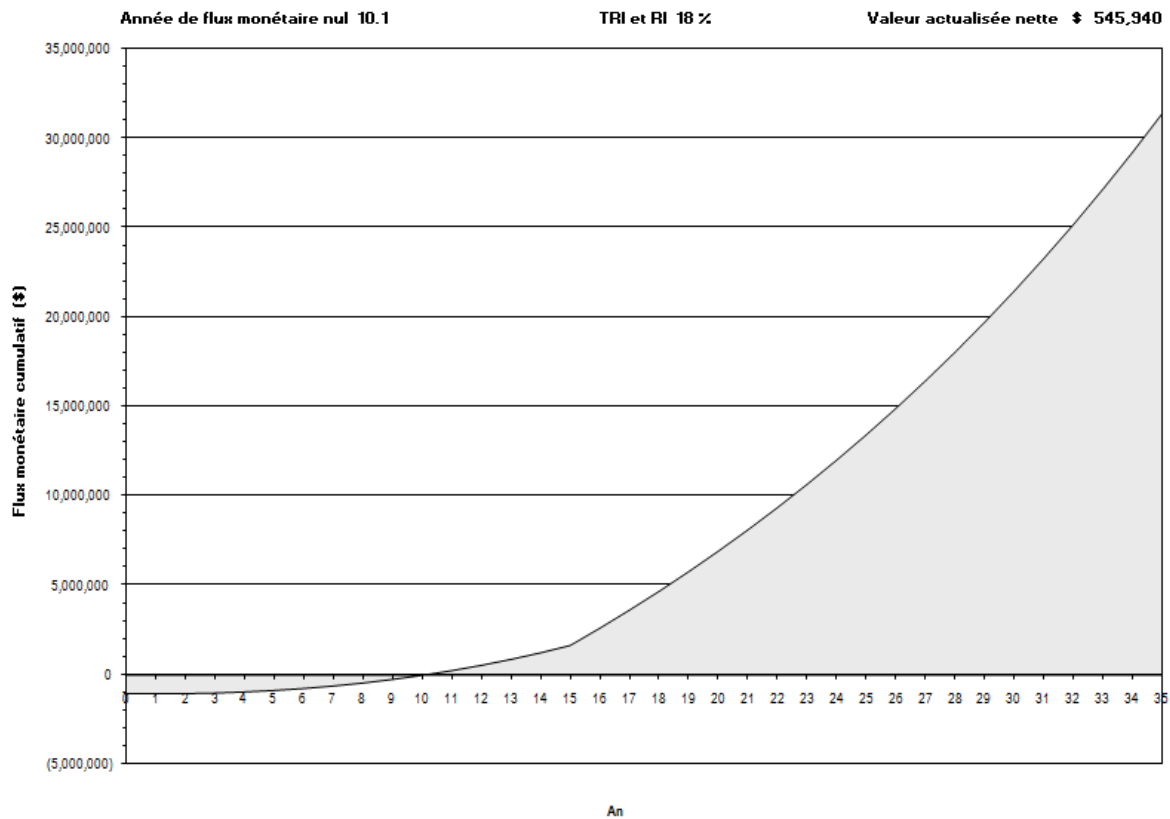


Figure39. Cumulative cash flow graph of Gisuma MHP

Table 10.Summary of important parameters values

S/N	Parameter	Value
1	Proposed Installed Capacity	451kW
2	Proposed Available Gross Head	131.5 m
3	Net Head of the Project	125m
4	Design Flow	0.561 m ³ /sec
5	Proposed length of the headrace channel	200m
6	Proposed length of penstock path	131.5m
7	Proposed average pipe diameter	510 mm
8	Penstock pipe material	Steel
9	Turbine Type	Horizontal Shaft Francis

10	No. of load centers	2
11	Total length of transmission line	0km
12	Total Length of Distribution Lines	2.5km
13	System Voltage	380V

Recommendations

Hydropower projects offer great opportunities for sustainable energy development of the developing countries. Being the cheapest, domestic, and renewable resource of energy, it deserves to be high up on the agenda of Rwanda Government's strategies in energy sector development. It is also advisable to create business opportunities for private companies in collaboration with rural people in that field. But, the tariff will be a challenge especially in the developing countries like Rwanda. To sort out this issue Rwanda Government will be advised to be involved in fixation of electricity costs, in rural areas, as it was done for other communities elsewhere , if not the rural citizens cannot afford to use the energy produced because of high price per unit (kWh).

Our recommendation was regarding what was not considering in this study, but need to be done later before implementation of this project model; the following are these points:

- The bills of quantities of any part and corresponding technical specifications for specific equipment;
- The different drawings regarding all the project;
- Dimensions of the proposed weir,
- The cost of all excavations needed from the weir to power house;

Also you should find the consumer load distribution off the domestics, business (shops), other like office, schools, hospitals, etc... in the appendix A1.

Conclusion

The choice of using micro-hydro power plant to supply electricity to the population of Muganza sector is motivated by several reasons. The Muganza population reside in rural areas and have no access to electricity, but there is a river of Gisuma as natural resource to be used in order to produce electricity.

Hydropower has a long service life, require basics skills but are work-intensive to maintain, and have a well developed service infrastructure.

Micro hydropower systems are however not simple to install and require larger investment depending to the location of river in which run off river system is used.

Micro hydropower plant has the advantage of producing electricity at low cost depending on the few parameters considered for hydropower plant construction and operation.

The model was used to calculate the firm capacity of 451kw at the site based on the firm flow, the gross head, design flow and efficiencies/losses .For off-grid applications, comparing the firm capacity with the peak load provides an indication of the additional power that is required from other sources (e.g. diesel generators).

To help Rwanda to meet the energy needs economically, environmentally and socially, sustainable ways while saving money and increasing energy security and self-reliance especially in rural sector, this GISUMA MHP case study is taken as one of the solutions to accelerate growth and development and poverty reduction in Rwanda.

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APPENDICES

Appendix A1: SAMPLE FORM SHEET FOR GISUMA MICRO-HYDROPOWER PLANT SURVEY.

1.GENERAL DATA OF REGION.

1.1	Province	South Province
1.2	District	Gisagara
1.3	Sector	Muganza
1.4	Cell	Rwamiko
1.5	Village	Kabacuzi

2. ACCESSIBILITY.

2.1	From District to Sector	
2.1.1	Distance	km
2.1.2	Road condition	Asphalt
2.1.3	Trip time	25minutes

2.1.4	Transportation mode	Public vehicle
2.2	From Sector to Cell	
2.2.1	Distance	km
2.2.2	Road condition	Asphalt
2.2.3	Trip time	25minutes
2.2.4	Transportation mode	Public vehicle
2.3	From Cell to Site location	
2.3.1	Distance	500m
2.3.2	Road condition	Rock
2.3.3	Trip time	5minutes
2.3.4	Transportation mode	Motocycle

3. DEMOGRAPHY LOCATION OF GISUMA RIVER

3.1	Total population	10015
3.2	Total house	2314
3.3	Population distribution	Grouping

4. VILLAGE INFRASTRUCTURE

4.1. PUBLIC FACILITY:

4.1.1	School	Primary school; Secondary school;
4.1.2	House of worship	Catholic Church
4.1.3	Health services	Local government: Byiza Health center
4.1.4	Government office	Kirarambogo Cell and Cyidatemwa cell

4.2: LIMIT OF VILLAGE

4.2.1	North	Gicumbi and Muganza Sector
4.2.2	South	Cymba and Soga Cell
4.2.3	West	Gishubi Sector
4.2.4	East	Ngozi District of Burundi Country

5. HYDROLOGY

5.1	Average measured discharge	...0.5632.....m ³ /s
5.2	Input discharge	...0.6..... m ³ /s
5.3	Geodetic head	131.....m
5.4	Hydropower potential	556. kw
5.5	Generated electric power potential	500..... kw

6. ELECTRIC POWER ENERGY DEMAND

6.1: Electric Power Energy Consumption

Let us consider 1000 house to be connected and 10 mill machines

6.1.1	Housing(per family head)	
	Illumination	4 lamp each 25w:4x25=100wx1000= 100kw
	Home appliances	Radio each 40w;TV each 100w; Toatal=140wx1000= 140kw
6.1.2	Productive Business	
		Mill machine :there are 10 each of 7.5kw Total=10x7.5= 75kw
6.1.3	Public facilities	
	Local government	10 lamps per cell and 3 computer : 0,350w
	5 Schools	There are 12 rooms each will have 4 lamps of 40w. Total=40wx4x12x5= 9,6w
	Estimated power consumption	=100+140+75+0,35+9,6= 225kw

7. RESOURCES

7.1	Construction Material	
7.1.1	Rock	Local Material shop.
7.1.2	Sand	Local Material shop
7.1.3	Others	Local area
7.2	Labours	
7.2.1	Availability	Carpentry Craft man are available
	Solary	monthly
7.3	Energy Sources and Utilization	
7.3.1	Existing generator	

	Hydro/turbine	Quantity Status Capacity: 0 kw
	Diesel	Quantity Status Capacity: 0 kw
	Solar Power	Quantity Status Capacity: 0,1 kw(at health center)
7.3.2	Illumination per house/family head	
	Petromax	Quantity oil consumption: l/day
	Wall lamp	Quantity oil consumption: l/day
7.3.3	Fire woods	Consumption : php/weak

QUESTIONNAIRE FOR HOUSEHOLDS OF NON-ELECTRIFIED VILLAGE

Household number	No number
Name of respondent	NZABANITA Emile
Name of village	CYADATUMWA
Interviewer's name	NZIRORERA Leonidas
Date	26/07/2013
Time interview began	13h20min
Time interview completed	13h40min

FAMILY PROFILE

1. Number of family members (only living together in the same house).

Male adults at 20 years or over3.....persons;

Female adults at 20 years or over4.....persons;

Children less than 20 years old3.....persons;
 Total10.....persons.

2. Number of schools

University student0.....persons;
 Secondary schools student1.....persons;
 Primary schools student2.....persons

3. How many of your family are earning income in the village? ...1.....persons;

4. How many of your family are living in other town to work?0.....persons;

5. How many of your family members graduated from high school?0...persons.

3. Housing

6. How many rooms does your house have?3 rooms;

7. What is floor area of your house?12...m²

4. Energy related properties.

8. Do you have following equipment for lighting?

Kind of equipment	Generator	Kerosene Lamp	Car battery	Gas fired cooking	Other
Number	no	no	no	no	

9. What kind of electrical appliances does your household currently use?

Equipment	Number
Bulb/Fluorescent lamp	0

TV set	0
Radio	1
Refrigerator	0
Air conditioners	0
Other	

4. NEEDS FOR ELECTRICITY

4.1: Priority Needs

10. Could you give your priority order on the following needs?

Water supply	5	
Education	2	
Health care	4	
Sanitation(toilet, solid waste, drainage)	3	
Electrification	1	
Public light on road	6	

4.2: Effort to have access to electricity.

11. Has your hold house over attempted to have access to electricity?

Answer is :No

12. What type of electricity generation did your household plan to have access to?

Diesel generator set

Solar home system

Wind power

Micro-hydropower :This is answer

Biomass

Other.....

6. PURPOSE OF USING ELECTRICITY.

13. If you can have access to electricity, what kind of electrical appliances and how many appliances do you want to use?

Equipment	Number
Bulb/Fluorescent lamp	4
TV set	1
Radio	1
Refrigerator	0
Air conditioners	0
Other	0

14. What public facilities do you think should have access to electricity?

School: This enserw

Mosque/Church;

Clinic/Health center;

Water pump for drinking water;

Others...

QUESTIONNAIRE FOR HOUSEHOLDS OF NON-ELECTRIFIED VILLAGE

Household number	No number
------------------	-----------

Name of respondent	KANAMUGIRE Etienne
Name of village	CYIRARAMBOGO
Interviewer's name	SHUMBUSHO Jean Pierre
Date	26/07/2013
Time interview began	15h05min
Time interview completed	15h28min

FAMILY PROFILE

1. Number of family members (only living together in the same house).

Male adults at 20 years or over2....persons;

Female adults at 20 years or over3.....persons;

Children less than 20 years old3.....persons;

Total8.....persons.

2. Number of schools

University student1.....persons;

Secondary schools student2.....persons;

Primary schools student0.....persons

3. How many of your family are earning income in the village?...2.....persons;

4. How many of your family are living in other town to work?.....0.....persons;

5. How many of your family members graduated from high school?....1...persons.

3. Housing

6. How many rooms does your house have?5 rooms;

7. What is floor area of your house?43...m²

4. Energy related properties.

8. Do you have following equipment for lighting?

Kind of equipment	Generator	Kerosene Lamp	Car battery	Gas fired cooking	Other
Number	1	3	1	no	

9. What kind of electrical appliances does your household currently use?

Equipment	Number
Bulb/Fluorescent lamp	12
TV set	1
Radio	2
Refrigerator	0
Air conditioners	0
Other	0

4. NEEDS FOR ELECTRICITY

4.1: Priority Needs

10. Could you give your priority order on the following needs?

Water supply	4	
Education	2	
Health care	3	

Sanitation(toilet, solid waste,drainage)	5	
Electrification	1	
Public light on road	6	

4.2: Effort to have access to electricity.

11. Has your hold house over attempted to have access to electricity?

Answer is :Yes

12. What type of electricity generation did your household plan to have access to?

Diesel generator set

Solar home system

Wind power

Micro-hydropower :This is answer

Biomass

Other.....

6. PURPOSE OF USING ELECTRICITY.

13. If you can have access to electricity, what kind of electrical appliances and how many appliances do you want to use?

Equipment	Number
Bulb/Fluorescent lamp	12
TV set	2
Radio	3
Refrigerator	1
Air conditioners	0

Other	0
-------	---

14. What public facilities do you think should have access to electricity?

School

Mosque/Church;

Clinic/Health center;

Water pump for drinking water;

Others...

QUESTIONNAIRE FOR HOUSEHOLDS OF NON-ELECTRIFIED VILLAGE

Household number	No number
Name of respondent	MUJAWIMANA Annonciate
Name of village	GATOVU
Interviewer's name	NZIRORERA Leonidas
Date	26/07/2013
Time interview began	15h23min
Time interview completed	15h42min

FAMILY PROFILE

1. Number of family members (only living together in the same house).

Male adults at 20 years or over1.....persons;

Female adults at 20 years or over1.....persons;

Children less than 20 years old0.....persons;

Total2.....persons.

2. Number of schools

University student0.....persons;

Secondary schools student0.....persons;

Primary schools student0.....persons

3. How many of your family are earning income in the village? ...1.....persons;

4. How many of your family are living in other town to work?0.....persons;

5. How many of your family members graduated from high school?0...persons.

3. Housing

6. How many rooms does your house have? ...4 rooms;

7. What is floor area of your house?25...m²

4. Energy related properties.

8. Do you have following equipment for lighting?

Kind of equipment	Generator	Kerosene Lamp	Car battery	Gas fired cooking	Other
Number	no	no	no	no	

9. What kind of electrical appliances does your household currently use?

Equipment	Number
Bulb/Fluorescent lamp	0

TV set	0
Radio	1
Refrigerator	0
Air conditioners	0
Other	

4. NEEDS FOR ELECTRICITY

4.1: Priority Needs

10. Could you give your priority order on the following needs?

Water supply	5	
Education	2	
Health care	4	
Sanitation(toilet, solid waste, drainage)	3	
Electrification	1	
Public light on road	6	

4.2: Effort to have access to electricity.

11. Has your hold house over attempted to have access to electricity?

Answer is :No

12. What type of electricity generation did your household plan to have access to?

Diesel generator set

Solar home system

Wind power

Micro-hydropower :This is answer

Biomass

Other.....

6. PURPOSE OF USING ELECTRICITY.

13. If you can have access to electricity, what kind of electrical appliances and how many appliances do you want to use?

Equipment	Number
Bulb/Fluorescent lamp	6
TV set	1
Radio	1
Refrigerator	0
Air conditioners	0
Other	0

14. What public facilities do you think should have access to electricity?

School:

Mosque/Church;

Clinic/Health center;

Water pump for drinking water;

Others...

QUESTIONNAIRE FOR HOUSEHOLDS OF NON-ELECTRIFIED VILLAGE

Household number	No number
Name of respondent	KANAMUGIRE
Name of village	CYIDATYURWA
Interviewer's name	SHUMBUSHO Jean Pierre
Date	26/07/2013
Time interview began	15h25min
Time interview completed	15h55min

FAMILY PROFILE

1. Number of family members (only living together in the same house).

Male adults at 20 years or over4.....persons;

Female adults at 20 years or over3.....persons;

Children less than 20 years old2.....persons;

Total9.....persons.

2. Number of schools

University student1.....persons;

Secondary schools student2.....persons;

Primary schools student2.....persons

3. How many of your family are earning income in the village? ...3.....persons;

4. How many of your family are living in other town to work?1.....persons;

5. How many of your family members graduated from high school?1...persons.

3. Housing

6. How many rooms does your house have?5 rooms;

7. What is floor area of your house?64...m²

4. Energy related properties.

8. Do you have following equipment for lighting?

Kind of equipment	Generator	Kerosene Lamp	Car battery	Gas fired cooking	Other
Number	1	3	1	no	

9. What kind of electrical appliances does your household currently use?

Equipment	Number
Bulb/Fluorescent lamp	6
TV set	2
Radio	3
Refrigerator	0
Air conditioners	0
Other	

4. NEEDS FOR ELECTRICITY

4.1: Priority Needs

10. Could you give your priority order on the following needs?

Water supply	6	
Education	2	

Health care	3	
Sanitation(toilet, solid waste,drainage)	4	
Electrification	1	
Public light on road	5	

4.2: Effort to have access to electricity.

11. Has your hold house over attempted to have access to electricity?

Answer is :No

12. What type of electricity generation did your household plan to have access to?

Diesel generator set

Solar home system

Wind power

Micro-hydropower :This is answer

Biomass

Other.....

6. PURPOSE OF USING ELECTRICITY.

13. If you can have access to electricity, what kind of electrical appliances and how many appliances do you want to use?

Equipment	Number
Bulb/Fluorescent lamp	12
TV set	2
Radio	3
Refrigerator	1

Air conditioners	0
Other	0

14. What public facilities do you think should have access to electricity?

School: 1

Mosque/Church;1

Clinic/Health center;1

Water pump for drinking water;

Others...

Appendix A2: Efficiency of Japanese generating equipment (TEPCO, 2005)

Efficiency of Japanese generating equipment

Output	Efficiency (%)		
	Turbine η_t	Generator η_g	Combined η_{tg}
- 100 kW	79	91	72
100 - 300 kW	81	93	75
300 - 1000 kW	83	94	78
1000 - 2500 kW	84	95	80
2500 - 5000 kW	85	96	82
5000 -10,000 kW	86	96	83

Appendix A3: Manning coefficient n for several commercial pipes (Penche, 2004).

Kind of pipe n	
Welded steel	0.012
Polyethylene (PE)	0.009
PVC	0.009
Asbestos cement	0.011
Ductile iron	0.015
Cast iron	0.014
Wood-stave (new)	0.012
Concrete (steel forms smooth finish)	0.014

Appendix A4: Kinematic Viscosity of Water in SI Units

Temperature - <i>t</i> - (°C)	Dynamic Viscosity - μ - (N s/m ²) x 10 ⁻³	Kinematic Viscosity - ν - (m ² /s) x 10 ⁻⁶
0	1.787	1.787
5	1.519	1.519
10	1.307	1.307
20	1.002	1.004
30	0.798	0.801
40	0.653	0.658
50	0.547	0.553
60	0.467	0.475
70	0.404	0.413
80	0.355	0.365
90	0.315	0.326
100	0.282	0.294

Appendix A5: RETScreen -Energy model(Site conditionss and System Characteristics)

Modèle énergétique RETScreen® - Projet de petite centrale hydroélectrique

Caractéristiques du site	Données	Notes/Plage
Nom du projet	feasibility study of gisuma hydro power plant(Rwanda)	
Lieu du projet	Gisuma-Muganza-Gisagara-Rwanda	
Hauteur de chute brute	m	131.5
Effet maximal de l'eau de fuite	m	2.00
Débit résiduel	m³/s	0.05
Débit garanti	m³/s	1.08
Charge de pointe du réseau électrique	kW	Réseau central
Demande du réseau électrique	MWh	Réseau central

Paramètres du système	Données	Notes/Plage
Type de réseau électrique	-	Réseau central
Débit nominal d'équipement	m³/s	0.960
Type de turbine	-	Francis
Nombre de turbines	turbine	1
Rendement de pointe de la turbine	%	91.7%
Rendement de la turbine au débit nominal	%	88.3%
Pertes hydrauliques maximales	%	5%
Rendement de la génératrice	%	95%
Pertes dans le transformateur	%	1%
Pertes parasites d'électricité	%	2%
Autres pertes et arrêts annuels	%	4%

Appendix A6: RETScreen - Equipment data (Small Hydro Turbine Characteristics)

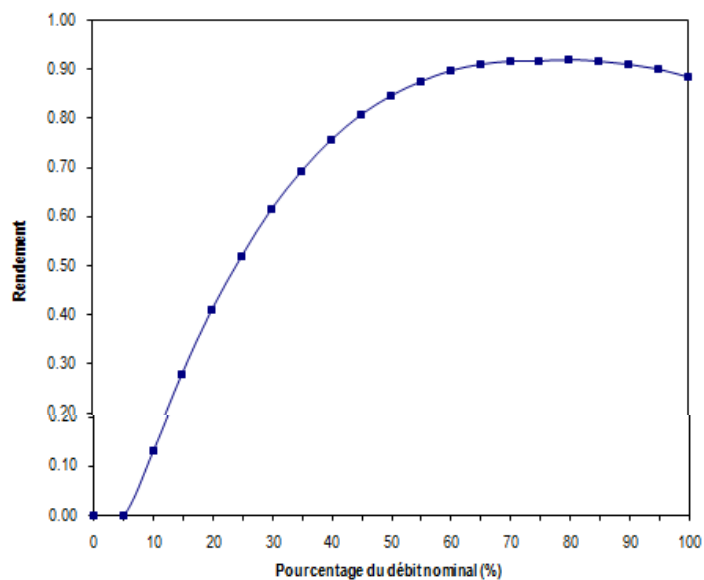
Caractéristiques des équipements RETScreen® - Projet de petite centrale hydroélectrique

Caractéristiques des turbines hydroélectriques		Données	Notes/Plage
Hauteur de chute brute	m	131.5	
Débit nominal d'équipement	m³/s	0.96	
Type de turbine	-	Francis	voir la base de données de produits
Source de la courbe de rendement de la turbine	-	Standard	
Nombre de turbines	turbine	1	
Manufacturier de turbines hydroélectriques		ABC Ltd.	
Modèle de turbine hydroélectrique		model XYZ	
Coefficient de conception de fabrication de turbines	-	4.5	2,8 à 6,1; par défaut = 4,5
Ajustement du rendement	%	0%	
Rendement de pointe de la turbine	%	91.7%	
Débit au rendement de pointe	m³/s	0.8	
Rendement de la turbine au débit nominal	%	88.3%	

Données de la courbe de rendement de la turbine

Débit (%)	Rendement de la turbine	Turbines en fonction #	Rendement combiné des turbines
0%	0.00	0	0.00
5%	0.00	1	0.00
10%	0.13	1	0.13
15%	0.28	1	0.28
20%	0.41	1	0.41
25%	0.52	1	0.52
30%	0.61	1	0.61
35%	0.69	1	0.69
40%	0.76	1	0.76
45%	0.81	1	0.81
50%	0.85	1	0.85
55%	0.87	1	0.87
60%	0.89	1	0.89
65%	0.91	1	0.91
70%	0.91	1	0.91
75%	0.92	1	0.92
80%	0.92	1	0.92
85%	0.91	1	0.91
90%	0.91	1	0.91
95%	0.90	1	0.90
100%	0.88	1	0.88

Courbe de rendement - 1 turbine(s)



Coûts d'investissement (crédits)	Unité	Quantité	Coût unitaire	Montant	% du total	Plage/quantité	Plage/coût
Étude de faisabilité							
Inspection du site	j-p	40	\$ 600	\$ 24,000	0.46%	-	-
Évaluation du potentiel hydraulique	j-p	20	\$ 500	\$ 10,000	0.19%	-	-
Évaluation environnementale	j-p	60	\$ 500	\$ 30,000	0.58%	-	-
Conception préliminaire	j-p	30	\$ 500	\$ 15,000	0.29%	-	-
Estimation détaillée des coûts	j-p	20	\$ 600	\$ 12,000	0.23%	-	-
Préparation du rapport	j-p	18	\$ 700	\$ 12,600	0.24%	-	-
Gestion du projet	j-p	12	\$ 700	\$ 8,400	0.16%	-	-
Voyages et hébergement	voyage-p	10	\$ 5,000	\$ 50,000	0.96%	-	-
Autres	Coût	0	\$ -	\$ -	0.00%	-	-
Crédits - Étude de faisabilité	Crédit	0	\$ -	\$ -	0.00%	-	-
Sous-total :				\$ 162,000	3.12%	-	-
Développement							
Négociation du CAÉ	j-p	20	\$ 1,000	\$ 20,000	0.38%	-	-
Approbations et permis	j-p	10	\$ 700	\$ 7,000	0.13%	-	-
Droits fonciers	site	1	\$ 50,000	\$ 50,000	0.96%	-	-
Arpentage	j-p	30	\$ 500	\$ 15,000	0.29%	-	-
Financement du projet	j-p	0	\$ -	\$ -	0.00%	-	-
Services comptables et juridiques	j-p	10	\$ 1,000	\$ 10,000	0.19%	-	-
Gestion du projet	année-p	0.5	\$ 130,000	\$ 65,000	1.25%	-	-
Voyages et hébergement	voyage-p	10	\$ 2,500	\$ 25,000	0.48%	-	-
Autres	Coût	0	\$ -	\$ -	0.00%	-	-
Crédits - Développement	Crédit	0	\$ -	\$ -	0.00%	-	-
Sous-total :				\$ 192,000	3.69%	-	-
Ingénierie							
Conception et appels d'offres	année-p	1.0	\$ 130,000	\$ 130,000	2.50%	-	-
Émission des contrats	j-p	25	\$ 700	\$ 17,500	0.34%	-	-
Surveillance des travaux	année-p	12.0	\$ 130,000	\$ 1,560,000	30.02%	-	-
Autres	Coût	0	\$ -	\$ -	0.00%	-	-
Crédits - Ingénierie	Crédit	0	\$ -	\$ -	0.00%	-	-
Sous-total :				\$ 1,707,500	32.86%	-	-
Équipements énergétiques							
Turbines/générateurs, contrôles	kW	958	\$ 500	\$ 478,996	9.22%	-	-
Installation des équipements	%	10%	\$ 478,996	\$ 47,900	0.92%	-	-
Transport	%	10%	\$ 478,996	\$ 47,900	0.92%	-	-
Autres	Coût	0	\$ -	\$ -	0.00%	-	-
Crédits - Équipements énergétiques	Crédit	0	\$ -	\$ -	0.00%	-	-
Sous-total :				\$ 574,795	11.06%	-	-
Infrastructures connexes							
Chemin d'accès	km	8.0	\$ 50,000	\$ 400,000	7.70%	-	-
Défrichement	ha	12.0	\$ 20,000	\$ 240,000	4.62%	-	-
Excavation de la terre	m²	200.0	\$ 10	\$ 2,000	0.04%	-	-
Excavation du roc	m²	10.0	\$ 60	\$ 600	0.01%	-	-
Barrage en béton	m²	1	\$ 1,000	\$ 1,000	0.02%	-	-
Barrage en gabions de bois	m²	0	\$ -	\$ -	0.00%	-	-
Barrage en remblais	m²	0	\$ -	\$ -	0.00%	-	-
Assèchement	%	10%	\$ 1,000	\$ 100	0.00%	-	-
Évacuateur de crues	m²	0	\$ -	\$ -	0.00%	-	-
Canal	m²	8	\$ 50	\$ 400	0.01%	-	-
Conduite d'amenée	m²	100	\$ 1,200	\$ 120,000	2.31%	-	-
Tunnel	m²	0	\$ 150	\$ -	0.00%	-	-
Pipeline/conduite forcée	kg	125	\$ 7	\$ 875	0.02%	-	-
Centrale - Génie civil	m²	100	\$ 1,200	\$ 120,000	2.31%	-	-
Échelle à poissons	m vertical	0.0	\$ -	\$ -	0.00%	-	-
Ligne et poste de raccordement	km	10.0	\$ 50,000	\$ 500,000	9.62%	-	-
Transport	%	4%	\$ 1,384,975	\$ 55,399	1.07%	-	-
Aute	Coût	0	\$ -	\$ -	0.00%	-	-
Crédits - Infrastructures connexes	Crédit	0	\$ -	\$ -	0.00%	-	-
Sous-total :				\$ 1,440,374	27.72%	-	-
Divers							
Équipements spéciaux	projet	0	\$ -	\$ -	0.00%	-	-
Frais généraux de l'entrepreneur	%	12%	\$ 1,440,374	\$ 172,845	3.33%	-	-
Formation	j-p	13	\$ 700	\$ 9,100	0.18%	-	-
Intérêts durant les travaux	%	8.0%	\$ 4,076,669	\$ 326,134	6.28%	-	-
Frais imprévus	%	15%	\$ 4,076,669	\$ 611,500	11.77%	-	-
Crédits - Divers	Crédit	0	\$ -	\$ -	0.00%	-	-
Sous-total :				\$ 1,119,579	21.55%	-	-
Total des coûts d'investissement				\$ 5,196,248	100%	-	-

Frais annuels (crédits)	Unité	Quantité	Coût unitaire	Montant	% du total	Plage/quantité	Plage/coût
Exploitation et entretien							
Location des terres	projet	1	\$ -	\$ -	0.0%	-	-
Taxes foncières	%	0.0%	\$ 5,196,248	\$ -	0.0%	-	-
Droits hydrauliques	kW	958	\$ -	\$ -	0.0%	-	-
Primes d'assurance	%	0.40%	\$ 5,196,248	\$ 20,785	13.1%	-	-
Entretien de la ligne de transport	%	2.0%	\$ 500,000	\$ 10,000	6.3%	-	-
Pièces de rechange	%	0.50%	\$ 5,196,248	\$ 25,981	16.4%	-	-
Main-d'oeuvre	année-p	2.00	\$ 35,000	\$ 70,000	44.2%	-	-
Voyages et hébergement	voyage-p	3	\$ 1,000	\$ 3,000	1.9%	-	-
Frais généraux et d'administration	%	11%	\$ 129,766	\$ 14,274	9.0%	-	-
Autres frais	Coût	0	\$ -	\$ -	0.0%	-	-
Frais imprévus	%	10%	\$ 144,041	\$ 14,404	9.1%	-	-
Total des frais annuels			\$	158,445	100%	-	-
Coûts périodiques (crédits)							
Mise à niveau de la turbine	Coût	20 ans	\$ -	\$ -	-	-	-
			\$ -	\$ -	-	-	-
			\$ -	\$ -	-	-	-
Valeur résiduelle du projet	Crédit	-	\$ -	\$ -	-	-	-

[aller à la feuille Analyse des GES](#)

Appendix A8: RETScree -Yearly Cash Flows

Flux monétaires annuels			
An #	Av. impôts \$	Apr. impôts \$	Cumulatif \$
0	(1,039,250)	(1,039,250)	(1,039,250)
1	(11,926)	(11,926)	(1,051,175)
2	10,662	10,662	(1,040,513)
3	34,214	34,214	(1,006,299)
4	58,770	58,770	(947,529)
5	84,373	84,373	(863,156)
6	111,066	111,066	(752,090)
7	138,893	138,893	(613,197)
8	167,902	167,902	(445,295)
9	198,143	198,143	(247,152)
10	229,665	229,665	(17,487)
11	262,522	262,522	245,035
12	296,770	296,770	541,805
13	332,466	332,466	874,271
14	369,669	369,669	1,243,940
15	408,443	408,443	1,652,383
16	964,564	964,564	2,616,947
17	1,006,674	1,006,674	3,623,621
18	1,050,558	1,050,558	4,674,178
19	1,096,287	1,096,287	5,770,465
20	1,143,938	1,143,938	6,914,403
21	1,193,590	1,193,590	8,107,992
22	1,245,325	1,245,325	9,353,317
23	1,299,230	1,299,230	10,652,547
24	1,355,393	1,355,393	12,007,940
25	1,413,907	1,413,907	13,421,847
26	1,474,870	1,474,870	14,896,717
27	1,538,381	1,538,381	16,435,097
28	1,604,545	1,604,545	18,039,643
29	1,673,472	1,673,472	19,713,115
30	1,745,275	1,745,275	21,458,389
31	1,820,071	1,820,071	23,278,460
32	1,897,984	1,897,984	25,176,444
33	1,979,140	1,979,140	27,155,584
34	2,063,675	2,063,675	29,219,259
35	2,151,724	2,151,724	31,370,983

Appendix A9: RETScreen –Financial Summary (Financial parameters, Project cost and Savings, Financial Feasibility)

Paramètres financiers						
Coût évité en énergie	\$/kWh	0.0800	Ratio d'endettement	%	80.0%	
Crédit pour ÉR fournie	\$/kWh	-	Taux d'intérêt de la dette	%	9.0%	
			Durée de l'emprunt	an	15	
Crédit - réduction d'émiss. GES		-	Analyse d'impôt sur le revenu?	oui/non	Non	
Coût évité en puissance	\$/kW-an	-				
Taux d'indexation de l'énergie	%	4.0%				
Taux d'inflation	%	2.5%				
Taux d'actualisation	%	15.0%				
Durée de vie du projet	an	35				
Coûts du projet et économies générées						
Coûts d'investissement			Frais annuels et dette			
Étude de faisabilité	3.1%	\$ 162,000	Exploitation et entretien	\$	158,445	
Développement	3.7%	\$ 192,000				
Ingénierie	32.9%	\$ 1,707,500	Paiements sur la dette - 15 ans	\$	515,713	
Équip. énergétiques	11.1%	\$ 574,795	Total des frais annuels	\$	674,157	
Infrastr. connexes	27.7%	\$ 1,440,374				
Divers	21.5%	\$ 1,119,579	Économies ou revenus annuels			
Investissement total	100.0%	\$ 5,196,248	Énergie	\$	640,570	
Encouragements/subventions	\$	-	Puissance	\$	-	
			Total des écon. annuelles	\$	640,570	
Coûts périodiques (crédits)						
Mise à niveau de la turbine	\$	-				
	\$	-				
	\$	-				
Valeur résiduelle - Crédit	\$	-				
Analyse financière						
TRI et RI avant impôts	%	18.0%	Calcul du coût de revient d'ÉR	oui/non	Non	
TRI et RI après impôts	%	18.0%	Calcul du coût de réduction de GES	oui/non	Non	
Retour simple	an	10.8	Capitaux propres investis	\$	1,039,250	
Année de flux monétaire nul	an	10.1	Dette du projet	\$	4,156,998	
Valeur actualisée nette (VAN)	\$	545,940	Paiements sur la dette	\$/an	515,713	
Écon. annuelles sur la durée de vie	\$	82,511	Recouvrement sur la dette	-	0.98	
Indice de rentabilité	-	0.53				